NIHR | National Institute for Health and Care Research

Summary

Reference Number	NIHR305240
Lead Applicant	Miss Hannah Barnsley
Research Title	Making Magnetic Resonance Imaging (MRI) safer: Developing educational resources for radiographers and patients to improve the communication of safety-critical information
Plain English Summary	**Why we are doing this research** MRI scanners are machines that use strong magnets and a type of energy (radiofrequency) to produce detailed pictures of the inside of the body. MRI is regularly used in healthcare. In the UK, 4.1 million MRI scans were taken from March 2022 to March 2023. As a diagnostic tool, MRI is invaluable. Yet, it comes with risks if the correct information is not given to the radiographers before the scan. The magnetic force of the machine can cause metal objects to move, sometimes quickly and dangerously causing harm. The magnetic force can also affect metal objects in the body, causing them to move or twist. This could injure a patient or damage the object, stopping it from working. The metal objects could be pacemakers, joint replacements (hip/knee) or metal pieces from old wounds. Burns can be caused by metal objects heating inside or outside of the body. For the safety of patients, radiographers should carefully screen every person who enters the MRI room. A patient may not have read or understood the information given to them. They may have forgotten their surgical history or be unconscious, leaving the potential for harm. Training for screening patients about to receive an MRI is not compulsory but should be addressed to stop any harm to patients. Also, there are few resources to teach patients what they should know about MRI screening. MRI scanning is safe but every person who enters the scanning room must understand the risk and what can be done to deal with any potential problem. **Aim** To increase patient and staff knowledge, and improve MRI safety. Online (eLearning) training for staff and an infographic and animation for patients will be made. Both will be available throughout the UK. **Our plans** The eLearning training will teach radiographers special screening methods to find and name medical devices on X-rays. In the UK, X-rays can be used for screening unconscious patients, but currently, there is no training for this. Also, if a patient already has X-rays, we could help patients with memory problems or communication difficulties. This research will test the program and find out if it changes how radiographers work. To teach patients about MRI screening, an information resource known as an infographic will be developed. It will explain what patients need to bring with them and the information patients need to tell the radiographer. Infographics are used in healthcare because they explain complicated information in an easy, visually inclusive manner. A web link will be added to an animation with subtitles. This allows anyone to listen to the same information. The infographic will be designed to be sent with appointment letters from any NHS hospital. The animations will be made available to all NHS Trusts who wants to use it on their website. Patients will help to design the infographic and animations. **Patient and Public Involvement (PPI)** Patients were asked their views on MRI screening. The feedback was talked about with the project's PPI team. They suggested using an infographic, with animations and subtitles, to help patients understand MRI safety screening. The PPI team helped shape all parts of this research plan from the design to finding people to take part, how to share the results and the plain English summary. **Sharing the findings** I will share the eLearning training and infographic/animation through MRI professional groups, social media, charities, and patient forums. I will do this with the co-creation, PPI and healthcare professional groups. Also, I will share my findings at patient events, presentations at MRI and radiology-specific conferences and articles in

	scientific journals.

Our purpose for collecting this data is to communicate with you about your application and have the necessary information to evaluate you for a grant. The data we collect here is collected in the public interest. Information provided here may be subject to Freedom of Information requests.

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1. Application Summary Information

Host organisation (which will administer any award)
The Royal Marsden NHS Foundation Trust

Partner organisation	Institute of Cancer Research: Royal Cancer Hospital (The)

Research Title	Making Magnetic Resonance Imaging (MRI) safer: Developing educational resources for radiographers and patients to improve the communication of safety-critical information

Partnership Fellowship Please select the relevant Partnership Fellowship.
N/A

Proposed WTE Please select which WTE option you would prefer
1.00

Duration (months) The duration for your award is based on your chosen WTE
36

Proposed start date if grant awarded	01/04/2025

Research Type	Other (please specify)

Please provide more details on your selection above.
Mixed-methods

2. Applicant CV

A CV for the Applicant is a mandatory component of this application; please ensure this is completed fully.
Please note that some of the CV questions are automatically completed from information you have provided in the 'Update CV' section of the 'Manage My Details' page. These can be updated via the ARAMIS Home Page or by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

Degrees and Professional Qualifications

From	To	Qualification	Subject	Country	Institution	Class
09/2016	11/2018	MSc	Magnetic Resonance Imaging (MRI)	United Kingdom (England)	Anglia Ruskin University	Distinction
09/2002	07/2005	BSc(Hons)	Diagnostic radiography	United Kingdom (England)	University of Portsmouth	2:2

Present and previous positions
(including from/to dates, job title and employer)

From	To	Position	Organisation	WTE % Research
09/2023	08/2024	NIHR Pre-Doctoral Clinical and Practitioner Academic Fellow	The Royal Marsden NHS Foundation Trust	100
01/2021	-	MRI Radiographer	The Royal Marsden NHS Foundation Trust	0
01/2021	-	Bank Neuroradiographer	King's College Hospital NHS Foundation Trust	0
06/2019	-	Emergency Medical Team Radiographer (volunteer)	UK-MED	0
03/2015	01/2021	Neuroradiographer	King's College Hospital NHS Foundation Trust	0
01/2015	03/2015	Locum Radiographer	East Surrey Hospital, Surrey and Sussex Healthcare NHS Trust	0
05/2014	09/2014	Locum Radiographer	East Surrey Hospital, Surrey and Sussex Healthcare NHS Trust	0
05/2011	01/2013	Radiographer	Lake Imaging, Ballarat. Australia	0
11/2010	01/2011	Volunteer Radiographer	Angkor Hospital for Children, Siem Reap. Cambodia	0
04/2008	06/2010	Radiographer	Queensland Diagnostic Imaging, Sunshine Coast and Brisbane. Australia	0
03/2007	04/2007	Locum Paediatric Radiographer	Royal Children's Hospital, Brisbane. Australia	0
03/2007	01/2008	Radiographer	Waitakere Hospital, Waitemata District Health Board, Auckland. New Zealand	0
08/2005	02/2007	Radiographer	Horton General Hospital, Oxford University Hospitals NHS Foundation Trust	0

Present and previous positions
(including from/to dates, job title and employer)

From	To	Position	Organisation	WTE % Research
07/2004	09/2004	Student radiographer	Queen Elizabeth Hospital, Kowloon. Hong Kong	0
09/2002	07/2005	Student radiographer	Queen Alexandra Hospital and Royal Hospital Haslar, Portsmouth Hospitals University NHS Trust	0
09/2002	07/2005	Student radiographer	Princess Royal Hospital, University Hospitals Sussex NHS Foundation Trust	0
09/2002	07/2005	Student radiographer	Dorset County General Hospital, Dorset County NHS Foundation Trust	0

Research Grants held

Publication Record
This should display the authors, date, article title and journal name for each publication added. This information can be edited via the “My Research Outputs” page of your ARAMIS account.

Publications
Barnsley, H., Robertson, S., Cruickshank, S. and McNair, HA.. (2024) ‘Radiographer training for screening of patients referred for Magnetic Resonance Imaging: A scoping review.’, *Radiography (London, England : 1995)*, 30(3).
Barnsley, H., Uzoukwu, S., Hassan, S. and Borri, M.. (2023) ‘The use of low dose CT scouts for MR safety screening: A multi-reader evaluation.’, *Radiography (London, England : 1995)*, 30(1).
Connor, SEJ., Borri, M., Pai, I. and **Barnsley, H.**. (2020) ‘'Black Bone' magnetic resonance imaging as a novel technique to aid the pre-operative planning of posterior tympanotomy for cochlear implantation.’, *Cochlear implants international*, 22(1).

Relevant Prizes, Awards and other Academic Distinctions
Please note the information for additional prizes or awards is automatically populated from information provided in the "Update CV" section of the "Manage My Details" page, which you can update by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

Date (mm/yyyy)	Source	Prize	Role of Applicant
01/09/2023	National Institute for Health and Care Research	I was awarded an HEE/NIHR ICA Pre-doctoral Clinical and Practitioner Academic Fellowship 2023 (NIHR303357)	Primary applicant
01/07/2022	The College of Radiographers	I was chosen for the Formal Radiographer Research Mentoring scheme, which aims to increase research capacity/quality of radiographers and support future research leaders. https://vimeo.com/857556845	Primary applicant/Mentee

ORCID iD 0000-0003-2668-1855

3. Applicant Research Background

Professional Background	Allied Health Professional

Profession (Allied Health Professional)	Radiographer (Diagnostic)

Do you have current / active professional registration?
This registration must be with one of the **ICA approved regulatory bodies**

Yes

Professional registration number/pin
Please supply a valid professional registration number/pin followed by the regulatory body (eg 12A3456B - NMC or PA12345 - HCPC)

RA48211 - HCPC

Please describe your research career to date

Clinical

I am a radiographer with nineteen years of experience in radiology departments in the UK and internationally. I have gained experience in a wide range of radiological modalities, including specialised cardiac Computed Tomography (CT), neuroimaging and rare interventional procedures. While working in neuroimaging, I specialised in Magnetic Resonance Imaging (MRI), completing my MSc in 2018. During this time, I frequently encountered patients (adult and paediatric) where routine MRI safety screening methods were impossible, due to a lack of mental and/or physical capacity. This could be the result of major trauma, neurological issues (e.g., stroke), neurosurgical conditions (e.g., brain tumours/surgery) or because they were unconscious due to mechanical ventilation. Consequently, I explored alternative screening methods. Upon leaving neuroimaging, I was struck by the multitude and diversity of medically implanted devices encountered in neuro MRI, uncommon in different population groups. This motivated me to investigate ways to enhance radiographers' awareness of medical implants, demonstrating my commitment to using research to inform clinical practice to improve patient safety and outcomes.

Research experience

For my MSc, I investigated utilising a novel MRI sequence named 'Black Bone' to avoid CT radiation for pre-operative planning for cochlear implant insertion. I used experimental methods to optimise the 'Black Bone' sequence using MRI phantoms and a within-person method for image acquisition and image quality analysis. Since publication (2020)[1], the 'Black Bone' protocol has been implemented at Guys and St Thomas's NHS Trust for patients undergoing cochlear implant insertion.

I then investigated whether CT scout images, acquired before a CT scan, could be used for MRI safety screening, using an analysis of repeatability between subjects method. UK guidance states that if a reliable history is unavailable, plain film radiographs could be taken to identify any MRI contraindications. CT scout images have quicker acquisition, lower radiation dose and the same anatomical appearance as radiographs. I had two abstracts accepted for poster presentations at a national (2021)[2] and international conference (2022)[3] and a manuscript, published in Radiography[4]. A feature article based on this work with CT scouts was also published by Aunt Minnie Europe[5], a radiology community website (https://www.auntminnieeurope.com/?sec=def) with a reach across Europe and beyond, further disseminating our innovation.

After engaging with colleagues and discussing my work, I conceived an idea for an MRI safety-based eLearning training program teaching radiographers to find and name medical implants using plain film radiographs (see project plan). Identifying implants will allow clinical decisions to be made on whether to proceed to scan and will allow greater utilisation of existing MRI safety policies. I have engaged with NHS England, who are keen to

collaborate and use the eLearning for Healthcare (eLfH) platform to host this nationally available, digital resource.

I was successful in being awarded an NIHR PCAF, during which:

- I undertook a scoping review to investigate MRI safety and screening and identified training gaps[6]. The subsequent manuscript is in press and has drawn interest from international MRI safety expert Frank Shellock (https://www.mrisafety.com/), who leads medical implant testing and MR categorisation. He hopes this proposed work can complement his PPI work regarding MRI safety and could provide a platform to disseminate my work to a global audience;
- I gained experience of PPI, engaging with the public using the NIHR Biomedical Research Centre's Cancer Patients' Voice platform (https://patients-voice.cancerbrc.org/the-pre-mri-screening-process-what-are-your-thoughts?tool=brainstormer#tool_tab). Patients provided great insights and valuable opinions regarding the MRI screening process and possible improvements. The responses highlighted common themes, which have been discussed and developed for this proposal with my project PPI team;
- I ran an MRI professional group meeting (managers, educators, safety experts and Society of Radiographers representatives). I aimed to understand the potential impact of my eLearning training on staff knowledge and competence and to identify any implementation barriers. Despite anticipating a positive impact on practice, the main barrier was considered the promotion of the programme to the MRI community.

Additionally, since being awarded the PCAF, I was invited to present at an NIHR fellowship information session for prospective applicants to share my experiences of applying for an NIHR fellowship (9/2023).

I have had an abstract promoting image review for routine MRI screening selected for an oral presentation at the UK Imaging and Oncology (UKIO) Conference in June (2024)[7]. I also co-authored another abstract, accepted for an ePoster presentation[8].

I have been invited to introduce the pilot training module at the MRI Safety Update and Innovations Day (https://www.mrisafetymatters.co.uk/mri-safety-update-nov-24-1), a hybrid, globally attended conference, in November (2024). The presentation will explain the programme's aims and highlight the benefits of image review in routine safety screening.

I have been invited to co-author a narrative review raising awareness of research in clinical practice and identifying key features of NIHR fellowships for radiographers, for publication in Radiography.

Participatory research

I was the radiographer expert for a PhD student investigating whole-body MRI for patients with melanoma, which has resulted in abstracts being accepted to conferences internationally[9,10,11,12].

Additional experience

- Since 2018, I have taught third-year radiography students CT head anatomy and pathology as a guest lecturer at London South Bank University;
- I successfully applied to two programmes to further my research skills. The College of Radiographers' Formal Radiographer Research Mentoring scheme (6/2022-6/2023) aimed to increase research capacity, quality, and leadership in radiography. The Steps to Fellowship Success scheme (9/2022-4/2023), delivered by our NIHR BRC and RDS London, helped facilitate competitive NIHR fellowship applications;
- I completed the Edward Jenner programme, hosted by the NHS Leadership Academy. I was awarded an NHS Academy Award in Leadership Foundations (12/2023).
- I was invited to present to the RMH medical physics department 'Implant recognition utilising existing imaging'. Medical physics assists radiographers in identifying implants and producing risk assessments if implant details cannot be obtained. However, the use of imaging by this group is under-utilised due to a lack of exposure reviewing medical images. Feedback suggested that, as non-clinical staff, in-person training to use imaging for implant identification was preferred.

I have gained some research experience; however, the PhD would allow me to develop and learn additional skills to further inform professional practice to benefit patients.

Have you already registered for a PhD?	No

Has this application, or a similar application, previously been submitted to this or any other funding body?
No

Contextual Factors
Please use this question to detail any contextual factors you wish to make the Selection Committee aware of. NIHR wants to know about any circumstances so that they may take them into consideration during the assessment of your application. Contextual factors may include:

- Career breaks due to parental leave, or periods of illness.
- Reduced time spent undertaking research due to a disability or caring responsibilities. This could include any physical or mental difficulty that may have impacted your research career. These are situations that have a significant impact on your ability to undertake research.
- Reduced opportunities to career support e.g. mentorship, and limited opportunities to undertake prior research and training.
- Impact of the COVID-19 pandemic on your research career.

Please also use this section to detail any other factors that may have impacted your research career not listed in the examples provided. The impact on your career to date will be specific to your particular circumstances but could include such impacts as limited opportunities to obtain grant funding or fewer publications. In general terms, mitigating factors should be significant, and relevant.

NIHR acknowledges that you may be reluctant, or uncomfortable disclosing relevant information that is sensitive. However, you should bear in mind that we are unable to take into account factors that you do not disclose. Please be assured that the information provided by you is sensitive and will be treated confidentially and in line with General Data and Protection Regulations (GDPR).

King's College Hospital:

While pursuing new research projects after my MSc, including researching NIHR fellowships, the Covid-19 pandemic started. Consequently, I could not pursue tasks outside my clinical role as I had neither the time nor the mental capacity to do so. However, as conferences went online, I took the opportunity to attend clinical and research-based lectures, listed below:

UK Imaging and Oncology (6/2020):

- Doctorate Den: Doctoral Supervision-notes to help you navigate the path
- Patient voice: Patient stories- improving care for the public and patients through patient engagement
- Artificial Intelligence in practice-a companion to radiology
- The new reality in MR

The British Institute of Radiology (7/2020):

- Patient-centred care in the era of artificial intelligence and technological innovation

Society of Radiographers (11/2020):

- Radiology Educators' webinar series: Achieving Excellence in Radiography Education and Research

The Royal Marsden:

Prior to my current role, and before being awarded an NIHR Pre-Doctoral Clinical Academic Fellowship, I had never worked in a department that championed radiographer research. As a full-time clinical radiographer, all my research (including my MSc) had been conducted on my own time and with limited career support. The support I did receive was informal and from outside of diagnostic radiography. In 2023 an NIHR diagnostic and therapeutic radiographer incubator was created. Led by my primary supervisor, the need to build research capacity, nationally, among radiographers has been recognised. I have received excellent support from a multidisciplinary team at RMH. However, support from within diagnostic radiography remains limited. RMH has a strong track record in

research doctorates with over 20 post-doctoral nurses and Allied Health Professionals (AHPs). Currently, eight nurses/AHPs are studying towards a doctorate and twelve are on a pre-doctoral pathway working towards a doctorate. Yet, radiographers remain underrepresented. One diagnostic and two therapeutic radiographers have completed PhDs with two more radiographers currently studying. I hope to build on this further with my colleagues as a champion for the benefits of research within clinical practice.

4. Plain English Summary of Research

Why we are doing this research

MRI scanners are machines that use strong magnets and a type of energy (radiofrequency) to produce detailed pictures of the inside of the body. MRI is regularly used in healthcare. In the UK, 4.1 million MRI scans were taken from March 2022 to March 2023. As a diagnostic tool, MRI is invaluable. Yet, it comes with risks if the correct information is not given to the radiographers before the scan. The magnetic force of the machine can cause metal objects to move, sometimes quickly and dangerously causing harm. The magnetic force can also affect metal objects in the body, causing them to move or twist. This could injure a patient or damage the object, stopping it from working. The metal objects could be pacemakers, joint replacements (hip/knee) or metal pieces from old wounds. Burns can be caused by metal objects heating inside or outside of the body.

For the safety of patients, radiographers should carefully screen every person who enters the MRI room. A patient may not have read or understood the information given to them. They may have forgotten their surgical history or be unconscious, leaving the potential for harm. Training for screening patients about to receive an MRI is not compulsory but should be addressed to stop any harm to patients. Also, there are few resources to teach patients what they should know about MRI screening. MRI scanning is safe but every person who enters the scanning room must understand the risk and what can be done to deal with any potential problem.

Aim

To increase patient and staff knowledge, and improve MRI safety. Online (eLearning) training for staff and an infographic and animation for patients will be made. Both will be available throughout the UK.

Our plans

The eLearning training will teach radiographers special screening methods to find and name medical devices on X-rays. In the UK, X-rays can be used for screening unconscious patients, but currently, there is no training for this. Also, if a patient already has X-rays, we could help patients with memory problems or communication difficulties. This research will test the program and find out if it changes how radiographers work.

To teach patients about MRI screening, an information resource known as an infographic will be developed. It will explain what patients need to bring with them and the information patients need to tell the radiographer. Infographics are used in healthcare because they explain complicated information in an easy, visually inclusive manner. A web link will be added to an animation with subtitles. This allows anyone to listen to the same information. The infographic will be designed to be sent with appointment letters from any NHS hospital. The animations will be made available to all NHS Trusts who wants to use it on their website. Patients will help to design the infographic and animations.

Patient and Public Involvement (PPI)

Patients were asked their views on MRI screening. The feedback was talked about with the project's PPI team. They suggested using an infographic, with animations and subtitles, to help patients understand MRI safety screening. The PPI team helped shape all parts of this research plan from the design to finding people to take part, how to share the results and the plain English summary.

Sharing the findings

I will share the eLearning training and infographic/animation through MRI professional groups, social media, charities, and patient forums. I will do this with the co-creation, PPI and healthcare professional groups. Also, I will share my findings at patient events, presentations at MRI and radiology-specific conferences and articles in scientific journals.

5. Scientific Abstract

Background

Magnetic Resonance Imaging (MRI) is increasingly utilised in healthcare, combining magnets and radiofrequency energy to produce cross-sectional images. However, these elements can pose significant risks to patients, potentially causing serious injury or death. Metallic objects can be turned into projectiles or medical implants inside the body can move/twist when exposed to the magnetic field. Metallic objects in and outside of the body can heat, causing burns. Consequently, rigorous screening is required to identify any MRI contraindications. Current screening methods (questionnaire and verbal review) can be ineffective, especially for patients with memory issues or communication difficulties. Radiographs can identify MRI contraindications, but no training exists to teach radiographers how to identify medical implants on imaging, potentially resulting in implants being missed/misidentified. Clinical consequences could include:

- Cancelled/delayed scans;
- Misidentified implants being wrongly introduced into MRI.

Furthermore, patients often do not read pre-scan information and there are few easily accessible resources to inform patients about MRI screening and the required documentation.

Question

Can bespoke educational resources improve radiographer and patient knowledge of safety-critical information before an MRI scan?

Aims

- To introduce an MRI training programme for radiographers, and assess its effectiveness on their knowledge, skills and confidence to make pre-MRI safety-critical decisions;
- To review and develop patient information to increase patient knowledge and assist radiographers during pre-MRI screening, reducing harm and improving safety.

Objectives

- Undertake a systematic review, identifying the barriers and enablers to eLearning implementation;
- Build an eLearning training program, teaching radiographers to identify implants using radiographs;
- Evaluate the changes in knowledge, skills and confidence of MRI radiographers pre- and post-training;
- Explore the impact of the training on MRI radiographers' awareness of their role in improving safety and reducing harm to patients;
- Develop complementary patient information to improve understanding of MRI screening.

Methods

The eLearning training will be built using the UK's Medical Research Council's framework for complex interventions. The evaluation will use a longitudinal concurrent mixed-methods approach, to understand the program's impact on radiographers, their professional practice and observed changes in patient safety.

A patient infographic and animation will be designed by co-creation utilising participatory action research.

Delivery timelines

Year 1: Ethics applications, the training build and systematic review will be completed;

Year 2: The training evaluation and infographic/animation development will start;

Year 3: Writing and submitting PhD thesis.

Before each objective commences, the required personal training will be completed.

Impact

Radiographers:

- Increased knowledge and confidence in safety-critical decision-making;
- Greater awareness of medical implants;
- Greater utility of existing MRI safety policies;
- Improving safety.

Patients:

- Allows time to recall their medical history;
- Faster access to MRI, especially in emergencies;
- Less risk of injury.

NHS:

- Fewer delayed/cancelled appointments, saving money and reducing waiting lists;
- Free training, reducing pressure on training budgets.

Dissemination

The eLearning training and infographic will be promoted through MRI professional organisations (e.g., the British Association of MR Radiographers), social media, charities and patient forums assisted by my co-creation, PPI and steering groups. Each aspect of the research will be published in scientific journals and presented at radiology-specific conferences.

6. Detailed Research Plan

1. **The problem**

Internationally, people are being seriously injured, and dying, every year because of improper practices and inadequate screening in MRI[13,14]. In 2001, 6-year-old Michael Colombini was killed during a routine post-operative MRI scan[15]. Since then, there has been a call for consistent policies on safety training and screening in MRI. However, although rare, serious harm to patients continues to be reported[14,16,17], with death occurring as recently as 2023[18]. The main causes are:

- **Radiographers not trained to screen patients for MRI[6];**
- **Patients are inadequately prepared for MRI safety screening.**

MRI uses powerful magnets and radiofrequency energy to generate detailed cross-sectional images. Ensuring safe entry into MRI is essential to prevent harm to patients. Current safety screening involves a written questionnaire followed by a verbal review with the patient, by MRI staff. The goal is to identify any medically implanted devices (e.g., pacemakers, joint replacements) or metallic foreign bodies (e.g., shrapnel). Identification is crucial as some may pose risks when exposed to MRI or require special scanning conditions.

Risks associated with introducing unknown/unsafe objects into MRI include:

- Ferromagnetic objects turning into projectiles (e.g., oxygen cylinders[19]);
- Moving/twisting of objects within the body, causing harm to patients or damage to the device (e.g., aneurysm clips[19]);
- Metallic objects inside or outside the body can heat, with burns being the most reported incident in MRI (41% of MRI incidents reported to the MHRA, 2017[17]).

A radiographer's inability to identify implants/foreign bodies can result in:

1. Introduction of unknown/unexpected implants into MRI;
2. Delays while implants are identified, particularly critical in emergencies;
3. Patients undergoing less suitable imaging, potentially exacerbating health inequalities based on clinical needs.

Radiographers are at the forefront of patient safety in MRI[14]. However, they are reliant on the information provided by patients being accurate. Patients lack awareness of the importance of MRI screening. This was confirmed by my PPI group, who despite having had MRIs, were surprised at how much information was needed and why. Patients can struggle to recall precise details of implants or complete surgical histories[17], leaving radiographers making safety-critical decisions based on potentially incomplete information. Raising awareness of MRI screening can be an opportunity to reduce adverse incidents. My PPI work demonstrated this could improve the patient experience.

Feedback from patients, when asked about MRI screening, included:

"Patients can feel anxious, which can give them brain fog or they may have cognitive issues. If the questions the radiographers need to ask just before the MRI could be explained on an information card and given to patients to read ahead of being verbally asked may be helpful."

"Pictures to depict the removal of all metal jewellery & clothing. Tell staff about implants so risk can be assessed."

"Explain in more detail pre-appointment what you want people to know/take?"

The exact number of MRI incidents and near-misses in the UK is unknown because MRI incident reporting is not mandatory[17,20]. However, I have experienced these issues and have presented case studies where patients either did not know they had an implant (high-risk pacemaker) or did not understand the type of implant in situ (MR Unsafe breast implant)[7]. This is reflected in a study conducted by a UK imaging provider[17], who, over three years, recorded incidents such as:

- Implants being mistakenly scanned outside of safety policies;
- Burns;
- Unexpected implants being introduced into MRI;

- Implant malfunction because of MRI.

Between March 2022-March 2023, 4.1 million MRI scans were performed in the UK (up 6.2% on 2021-2022). With an MRI safety incident rate of 1/1987 patients[17], it could be assumed over 2000 MRI safety incidents occur every year, ranging in severity.

This research will educate radiographers and patients to raise awareness and promote MRI safety. A national eLearning training program will teach radiographers to identify medical implants on radiographs. Currently only recommended for unconscious patients, incorporating reviewing pre-existing imaging into MRI screening could reduce the number of missed/unidentified implants introduced to MRI. Additionally, a co-created infographic and animation resource will educate patients about MRI screening, what information is required and its importance.

2. **Why is this research important?**

Gaps in the MRI screening process were identified in the preparatory work for this application. Shortfalls in radiographer education were highlighted through my scoping review[6] and patient feedback stressed a lack of accessible information regarding screening requirements.

Radiographers:

- To provide training in advanced screening methods, which currently does not exist;
- To increase radiographer knowledge and confidence in clinical decision-making, with lower confidence levels being recorded in radiographers without formal MRI education[22];
- Provides a greater awareness of medical implants and their purposes;
- Globally, radiology is severely understaffed. Consequently, radiographers' attention is often divided and their concentration reduced[14,23]. This research will facilitate faster, more robust screening and allow radiographers to better utilise MRI safety policies, if available.

Feedback from the professional group meeting included:

"Until you know about them, you don't look for them […] It's a library and not everyone has the opportunity to see this variety and breadth of implants"

Member8

"People specialise so early now and don't have the same plain film experience. This will help fill that knowledge gap"

Member9

Patients:

- To improve how patients receive information before scanning appointments. This may help patients feel better prepared and minimise an already stressful process. Better health literacy can enhance patient experience and improve safety by allowing patients to provide more accurate information;
- The absence of patient education increases the likelihood of missed implants entering MRI or cancelled/delayed scans. Disclosing implants at the appointment without the required documentation can delay imaging, especially critical due to long waits for MRIs (up to 2.5 years in some Trusts[21]). This research will increase MRI utility, reducing the use of less desirable modalities e.g., CT in time-sensitive situations;
- To facilitate faster access to MRI, especially in emergencies, allowing treatments or interventions to be enacted quicker;
- To ensure patient safety, reducing injuries, especially for those who cannot recall, or have complex medical histories.

"It helps with patient management, patient flow, patient waiting times and reduces delays in care"

Member7

NHS:

- Fewer delayed/cancelled appointments due to unknown/unidentified implants, saving money and reducing waiting lists (it can cost up to £1000 per missed MRI appointment that cannot be filled);
- Free access, relieving pressure on training budgets.

"The eLearning for Healthcare platform is designed to go onto hospital sites. If you do online learning, this training could be integrated into it"

Member2

"It's free at the point of access. Any healthcare professional can access it"

Member8

Considering the PPI feedback and highlighted training gaps, compounded with early career specialisation, chronic staff shortages[24] and a continued increase in MRI activity[25,26], there's a potential for increased MRI safety incidents over time.

3. **Review of existing evidence**

Radiographers:

In the UK, MRI is taught at the undergraduate level, but there is no evidence that core skills can be sufficiently attained due to minimal hands-on experience and competing curriculum requirements[6]. Research indicates that radiographers with formal MRI education have more confidence, knowledge and clinical decision-making skills than those without[22], with negative correlations recorded between MRI safety knowledge and safety incident numbers[13,22]. The UK has post-graduate MRI education, but current NHS finances and staffing often prevents radiographers from participating. This highlights the need for high-quality, free and easily accessible training for radiographers to promote MRI safety.

Basic MRI safety training is mandatory for anyone to work in MRI, often delivered locally in varying formats (online/face-to-face). However, it is also delivered nationally via eLearning (https://portal.e-lfh.org.uk/Component/Details/455458). The success of this standardised training has seen Trusts embed this within their local training programs as different modules are structured towards different responsibilities in MRI. Despite a lack of formal evaluations for eLearning programs, studies have shown success when utilising purpose-built, interactive sessions[27].

My scoping review[6] identified that radiographers are not taught to screen patients and no training exists to identify medical implants or extra-ocular foreign bodies on imaging. If a patient cannot answer a safety questionnaire (e.g., unconscious/unresponsive patients) and there are no up-to-date medical records/next of kin, the MHRA advises using radiographs and physical examinations to exclude any MRI contraindications. Physical examinations are becoming less sensitive with technological evolutions making implants harder to externally locate. E.g., traditional pacemakers (high-risk in MRI) implant a subcutaneous generator box, which leaves a scar. However, new 'leadless' pacemakers are implanted endovascularly leaving no scar, limiting the effectiveness of physical examinations.

Routine use of imaging for MRI screening is limited to localising metallic intra-ocular foreign bodies, which have caused severe injury in MRI, including blindness[28]. Radiographer reporting is an established role, but this does not include identifying implants on radiographs. There is an argument that if medical images are available, they should be routinely used for MRI screening for all patients[7], not just for unconscious/unresponsive patients. We evaluate images faster than text because images are processed as a whole whereas text is processed in lines[29]. Thus, incorporating image review into MRI screening can speed up the process, with medical records only used to confirm implant types. Additionally, using pre-existing imaging could increase radiographers' confidence in safety-critical decision-making, reducing reliance on patient information.

A study investigating radiographers' abilities to detect intra-ocular foreign bodies on radiographs[28] showed high sensitivity/specificity post-training, indicating that education regarding implant identification could positively impact safety-critical decision-making. Training to improve and standardise safety-critical decision-making has proven effective in other areas of healthcare. For example, the Systematic Anti-Cancer Therapy passport educational tool was introduced to combat the variation of skills and competencies of nurses administering chemotherapy[30]. It was

recognised that non-standardised training and increasing patient numbers with poor patient education resulted in increased incidents, treatment delays and poor provider-patient relationships/patient outcomes. The introduction of the training has shown a positive impact on staff skills and competencies, better use of resources and improved patient experience and outcomes[31].

To address the gap in the MRI screening process, I have developed the pilot module of an eLearning training program to educate radiographers to identify medical implants on radiographs, using the Medical Research Council's Framework (MRCF) for designing complex interventions[32,33]. This training will enhance radiographers' skills and competencies using imaging to improve patient safety, especially for those with communication difficulties or memory impairment.

Phase 1 (MRCF[32,33]):

I constructed the initial concept and drafted a training prototype. Using an iterative process this was shared with stakeholders (radiographers and managers) to obtain feedback on MRI safety issues and elements unique to eLearning programs[34]:

- Content design: information to be included/excluded;
- Layout;
- Language;
- Implants to be included.

The pilot (chest) module is the first of four being built on our collaborator's platform, eLearning for Healthcare (eLfH), which allows free, unrestricted access to NHS staff. Each implant's name, function and appearance are described, with arrows indicating different elements and locations of each implant (Appendix 2). Five multiple-choice self-assessment questions follow the training.

Phase 2 (MRCF[32,33]):

A professional group meeting was held in March 2024. Asked to provide feedback on the pilot module, they predicted a positive response to the training and a positive impact on radiographers' practice. However, the promotion and engagement with the training were thought to be the largest barrier to implementation (Package 1).

Between the PCAF and DCAF awards, the module will undergo pilot testing using a voluntary, anonymous pre- and post-training questionnaire (Appendix 3). The results will be analysed at the start of the DCAF (Package 2) and will inform the development of the following three modules (abdomen&pelvis/head&neck/extremities). eLfH modules undergo regular review, allowing updates to the chest module. The second module (abdomen&pelvis) components will be collated before the DCAF (June 2024-March 2025) and reviewed when the pilot results are available.

Patients:

Online searches during my PCAF produced numerous resources explaining the MRI process, including easy-read materials, videos, and blogs. These resources explain what to expect from arrival until after the scan. However, they commonly lack specific information about screening, including what information will be asked. Consequently, patients arrive prepared to remove jewellery or anticipate the scanner's loud noise but are unprepared to answer screening questions, including providing medical implant documentation. Pre-existing imaging could be utilised to ensure there are no MRI contraindication[7], but this may not exist. Acquiring images using ionising radiation for MRI screening is generally reserved for emergencies. Patient feedback uncovered large gaps in awareness of MRI screening and the requirements, highlighting the need for educational resources.

4. **Research question:**

Can bespoke educational resources improve radiographers' and patients' knowledge of safety-critical information before an MRI scan?

Aims:

- To introduce an MRI training programme for radiographers, and assess its effectiveness on their knowledge, skills and confidence to make pre-MRI safety-critical decisions;
- To review and develop patient information to increase patient knowledge and assist radiographers during pre-MRI screening, reducing harm and improving safety.

Objectives:

- Undertake a systematic review, identifying the barriers and enablers to eLearning implementation;
- Build an eLearning training program, teaching radiographers to identify implants using radiographs;
- Evaluate the changes in knowledge, skills and confidence of MRI radiographers pre- and post-training;
- Explore the impact of the training on MRI radiographers' awareness of their role in improving safety and reducing harm to patients;
- Develop complementary patient information to improve understanding of MRI screening.

5. **Project Plan:**

Package 1: Systematic review

The systematic review aims to identify the barriers and enablers for the implementation of eLearning training and how to engage people with eLearning programs successfully. It is unknown how willing and able radiographers are to undertake additional training/continuing professional development activities, especially with limited opportunities during work hours due to poor staffing and large workloads. The professional group meeting identified the promotion and dissemination of the training to be the largest barrier to its implementation into practice. The results of this review will inform how I advertise and promote the program to the wider radiographic community who will most benefit from it and how best to engage MRI radiographers and managers in non-compulsory eLearning activities.

Study design

The review will be registered on the PROSPERO database and will follow the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) 2020[35].

Data collection

The research question and protocol, including the search strategy, will be agreed upon by the review team. Literature will be identified through electronic databases (e.g., PubMed), hand-searching relevant journals (e.g., E-Learning and Digital Media) and reference lists, assisted by a research librarian. Inclusion criteria will be based on the PICO framework.

Data analysis

A data extraction tool will be co-designed with my steering group. I will screen all identified literature for eligibility and perform data extraction and quality assessments with a colleague experienced in systematic reviews. A third party will resolve any disagreements between the reviewers.

Package 2: Building the intervention.

The eLearning training development uses the Medical Research Council's Framework (MRCF) for designing complex interventions[32,33], which offers an iterative, phased approach to build ideas[33]. This is underpinned by Theory of Change (ToC)[36], which facilitates understanding of how and why an intervention might work by constructing a causal pathway, illustrating how different elements can affect change. ToC can also integrate behavioural change theories to explain why connections happen[36]. This approach is essential for the proposed project because it enables mapping of short-to-long-term impacts and incorporates stakeholder input to tailor the program to the target population. This increases the likelihood of embedding it into routine practice and enhancing

patient safety.

In year one, the pilot module results will be analysed (see existing evidence). Feedback is aligned with the Kirkpatrick Model of learning[37], used to assess the effectiveness of Massive Open Online Courses, concentrating on first and second-level criteria (Reaction and Learning). Questions cover demographics, objective achievement, course material, confidence in applying the learning, and changes in attitude towards image review for MRI screening. Data collection will last three months pre-award (Appendix 3) or until receipt of 50 responses. The pilot will help determine recruitment feasibility for the main evaluation sample size (Package 3), recruitment strategies, and training design. Descriptive analysis with confidence intervals will be conducted for formative evaluation. Modifications to improve performance will be made in collaboration with the steering group. Short-term outcomes (Kirkpatrick Level 2: knowledge/skill acquisition[36]) and potential implementation barriers will be measured using Likert-style responses. A progression criterion of >70% has been set[38]; requiring >70% of participants to record improved knowledge and confidence post-training to proceed with the remaining modules. If the threshold is not met, the training will be redesigned using the feedback and the steering group.

Package 3: Intervention evaluation.

Phase 3: Evaluation[32,33]

ToC identified radiographers as key to implementing the training, and routine image review, into practice. Using the COM-B behaviour change wheel[38], issues that could affect radiographers' acceptance of the program include:

- Capability: user learning/attainment and confidence;
- Opportunity: programme accessibility;
- Motivation: attitudes towards image review/change in personal practice.

This package will use a within-person, longitudinal concurrent mixed-methods approach, collecting qualitative and quantitative data using Kirkpatrick's evaluation levels 2-4 (Learning, Behaviour and Results). It will assess the effectiveness of the completed intervention, participants' experiences and reasons for success/failure[40].

Inclusion

- Diagnostic and therapeutic MRI radiographers, Band 5-8a, as these grades are responsible for MRI screening but vary in their exposure to radiographic interpretation and experience.

Exclusion

- Medical physics, as they lack radiographic interpretation knowledge and experience;
- Participants without access to eLfH.

Recruitment

Participants will be recruited through the Society of Radiographers and social media platforms, assisted by the steering group. Radiographers undertaking post-graduate MRI education and teaching hospitals could be targeted. I will also submit a proposal to the Research Hub of UKIO 2026, a dedicated space to recruit participants from across the country. Gift vouchers will be used to promote interaction, helping to reduce self-selection bias. Recruitment strategies will be refined based on the pilot recruitment.

All data collection will be performed on the GDPR-compliant Collective Minds Radiology (CMRAD) online image storage platform (https://www.cmrad.com/). CMRAD allows participants, in different geographical locations, access to anonymised images and questionnaires while providing all the expected functions of a hospital-based system. Participants will access identical but randomised ordered radiographs, selected using pre-determined criteria e.g., good diagnostic quality, to be agreed with the steering group. Implant information will be confirmed through medical records. Sampling bias of images will be reduced by ensuring a wide variety of implants, ranging in conspicuity (Appendix 2)[28].

Sample size

100 recruited radiographers should enable the detection of a standardised effect size as small as 0.2 (α=0.05; power=80%) allowing us to detect small but meaningful changes between the mean scores of different time points. A smaller sample (n=50) could detect a standardised effect size of 0.4, assuming the same statistical error rates. These figures will be reviewed after the pilot study, as we can assess features such as recruitment strategies and dropout rates pre-and post-training. Calculations were performed in nQuery Advisor (Version 8)[41].

Data collection and analysis

Quantitative

Using Kirkpatrick Level 2 evaluation[37], user learning and confidence will be measured.

Participants will mark and name every implant (if any) within an image using a circular Region of Interest (ROI) and rate their confidence of an implant being present within each ROI using a confidence index score (1-5). Each image will have a total score of 1 assigned to it. For images with multiple implants, fractional scores will be given e.g., for two implants, a score of 0.5 per implant will be assigned. All implants are weighted equally, as each requires independent clinical decisions. Acceptance radiuses of the ROIs, which represent the maximum inaccuracy with no clinical impact, vary in the literature and will be determined post-radiograph selection[30]. Implant sizes will vary considerably, so thresholds must account for this.

Participants will have one month to complete all four training modules. Participants will test their knowledge before and immediately after the training, to determine if it improves diagnostic performance. A further test will be undertaken six months post-training to determine if participants have retained their knowledge.

I will perform the statistical analyses, assisted by a statistician. Descriptive statistics can be performed (e.g., sensitivity/specificity) to summarise the data. Statistical differences in mean results at each timepoint will be determined using a paired t-test for normalised data. A Wilcoxen signed-rank test will be used for non-normally distributed data.

Qualitative

Questionnaires will include open and closed questions to understand radiographers' attitudes toward image review, changes in safety-critical decision-making and the programme's influence on individual practice, incorporating elements of the COM-B behaviour wheel[39] and Kirkpatrick's levels 3+4[37]. EDI data will be collected from each participant.

Closed-ended questions will provide descriptive statistical data, contextualising quantitative results and targeting future program enhancements. Open-ended questions will undergo qualitative content analysis, utilising the Framework Method[42]. Coding will use a deductive approach due to the study's specific interests. However, 5-10 questionnaires will be open-coded to ensure important data is not excluded[42]. Pre-defined codes will be agreed with the steering group. Two reviewers (myself, and an experienced supervisor) will perform coding. The first few questionnaires will be coded independently allowing codes to be compared and discussed so they can be applied to the rest of the responses. This process will be repeated for each data collection time point.

Package 4: Patient intervention

The preparatory work for this project identified patient knowledge of MRI screening as an area for improvement. This package follows the co-creation framework by Leask, et al.[43].

Aim

To create an infographic accompaniment for patient appointment information to support improved MRI safety screening. This will be produced in both static printable and dynamic animated formats to support diverse needs. An infographic was chosen for its ability to present complex information simply, in a visually appealing manner[44]. Studies show that infographics can enhance comprehension, resulting in increased self-efficacy[45]. These resources will be open access to any Trust who wants to utilise them.

Design

Participatory Action Research (PAR) engages stakeholders in active learning and collective action to enhance a specific situation. PAR looks to locally produce high-quality, impactful material that strives to scale up to a population level[46].

The steps include:

1. Evaluating current healthcare infographics for effectiveness in behaviour change;
2. Discuss content and visual presentation;
3. Develop prototypes for visualisation and improvement[43];
4. Field-test prototypes for performance assessment outside meetings[43].

The process is expected to take 12-14 months. I will be supported by the Health Services Research Group within RMH, who have experience running co-production activities and a professional graphics/animation team who will develop our ideas.

(End-) Users

Patients attending for outpatient MRI scans.

Co-creators

Before co-creation begins, I will run two online focus groups for 10 patients, then radiographers, with a PPI co-facilitator. Per the evidence-based co-design methodology[47], each group will be questioned to identify each group's priorities. Semi-structured questions and consents will be developed with the PPI group. Meetings will be audio-recorded and transcribed. I will perform content analysis to highlight key themes to present to the co-creation team. Participants will be recruited through the NIHR's People in Research website (https://www.peopleinresearch.org/) and radiographers from London-based NHS Trusts.

A co-creation group of 10-12 members will be established, allowing for participatory dropout. Recruitment will be from Southeast London, reflecting the diverse mix of ethnicities, literacy levels, and socioeconomic backgrounds within the local population. Impactful infographics need co-creators to be representative of the end user population[41]. Recruitment (posters, video and easy-read formats) will use NHS PPI departments, advertisements in MRI departments, and targeting individuals from under-served groups[47] (e.g., local specialist education centres). EDI data will be collected from individuals expressing interest to ensure I include as many under-served groups as possible[49]. Convenience sampling (e.g., art collages) will also be utilised.

The co-creation process will take up to 30 hours[43]. My PPI team suggested meetings be summarised into bullet points and verbally recapped at subsequent meetings. Co-creation training, tailored to the group, will be offered before the first meeting, delivered by Arts and Homelessness International.

Acceptability review

Pre-co-creation patient and radiographer focus group members will be invited back to provide their opinions on the end infographic/animation products. Open questions will explore:

- What are their views and opinions of the final infographic/animation resources?
- Ease of use: how and where would the resources be most effective in the patient pathway?
- Implementation barriers: what would prevent the resources from being used? What would help the resources to be used? Would and how would the resources aid radiographic practice?

If the original focus group members are unavailable, the same recruitment strategies will be employed to ensure 10 participants in each group. The meetings will be audio-recorded and transcribed to allow content analysis. The results will inform and facilitate the design of the full-scale evaluation in my future work.

5. **Dissemination and outputs:**

The main intervention outputs of this award will be the eLearning training program for radiographers and the patient educational infographic, which will be held on the Society of Radiographers website and accessible to all radiographers.

I will disseminate these interventions through:

- Local Trust research forums;
- Academic publications;
- Conference presentations/workshops;
- Cancer Patients' Voice platform;
- Social media platforms;
- Imaging training academies (regional training and educational hubs https://www.hee.nhs.uk/our-work/cancer-diagnostics/training-academies/imaging-training-academies).

PPI dissemination:

- Patient forums;
- Charity affiliations;
- Community spaces;
- Social media platforms.

Professional steering group dissemination:

- Professional magazines and the Society of Radiographers website;
- Social media platforms;
- MRI safety-specific conferences and organisations;
- UK radiography-focused podcasts.

Additional plans regarding the dissemination of the training program and engagement with radiographers will be made after the systematic review. Clinical radiographers may not engage with scientific papers or attend conferences regularly.

Future funding:

- I will explore using machine learning to identify medical implants on radiographs. While MRI staff will still be responsible for naming implants, AI can assist by highlighting any foreign objects within an image, reducing the likelihood of implants being missed and decreasing the time needed for image review. AI is being utilised for image reporting with success, yet investigations into implant identification are not well-discussed. This can be explored through post-doctoral research funding, in collaboration with an industry partner.
- The infographic/animation resources will undergo a full evaluation to understand if they can impact patient behaviour, increase awareness of MRI safety and reduce imaging delays. Also, it will be translated into different languages assisted by post-doctoral research funding or the College of Radiographers Industry Partnership Scheme research grant.

Barriers to implementation:

Identifying barriers to eLearning implementation will be fully explored in the systematic review, the results of which will inform the implementation strategies.

Implementation barriers regarding the infographic/animation will be explored with the professional steering group and the patient/radiographer focus groups. Informatic Merseyside will assist with the technical aspects of implementation.

Intellectual Property/licensing:

There are no encumbrances/restrictions on any Background IP and the Foreground IP and Research Data will be owned by RMH. The expected foreground IP assets will include the training programme outlined above and related infographics and animations. The current plan is to license the training programme non-exclusively to appropriate parties to train other radiographers e.g., to the Society of Radiographers and/or NHS England for use

on the NHS England's learning platform (eLearning for Healthcare) and/or other similar training organisations. The NHS England platform has the potential to reach over 40,000 radiographers in the NHS. There is also potential to use them as part of the RMH's School (The Royal Marsden School | The Royal Marsden). The infographic and animation are likely to be made available on an open-access basis to ensure the greatest impact and use in the NHS. The research team will obtain IP and contractual support from the RMH Senior IP and Commercial Partnerships Manager and its legal team to ensure the appropriate contracts/licenses are implemented.

Anticipated impact:

Short-to-medium term:

Radiographers:

- Improved radiographer education;
- Improved confidence and safety-critical decision-making.

Patients:

- Better scan preparation and education;
- Safer and quicker access to MRI, especially in emergencies, enhancing health and well-being.

Long term:

NHS:

- Fewer missed/cancelled appointments, positively impacting finances and waiting lists;
- Faster screening in emergencies allowing treatments and interventions to be enacted quicker;
- Fewer adverse safety events, potentially reducing costs associated with additional medical treatments and potential litigation;
- Potential update to MHRA MRI guidelines to recommend image review as a part of routine MRI screening.

Project Management:

I will oversee the day-to-day management of the award, adhering to the research timetable (Appendix 3). This includes:

- Ensuring all ethical approvals and local permissions are in place;
- Recruitment and scheduling of PPI and steering group activities;
- Recruiting and consenting research participants;
- Managing data collection and analysis;
- Disseminating results;
- Meeting all PhD milestones.

This will be supported by my supervisory team, PPI and steering groups, with meetings scheduled per the research timetable.

The financial management of the award will be undertaken by the host organisation (The Royal Marsden).

Ethics:

An Equality Impact Assessment for this project found no identified barriers or impacts. However, scheduling of PPI activities will consider religious and school holidays, especially during the co-creation process.

Permission to use images for the eLearning training has been obtained from the Caldicot Guardians of both Trusts involved.

My research protocol will be submitted for local NHS and Health Research Authority approvals.

No physical or psychological harm is anticipated during the project. If distress does occur (e.g., re-living a traumatic scan experience), participants can pause or end their participation and I will assist in seeking out relevant support services.

Success Criteria:

Success measurements:

- The development and publication of the MRI eLearning training program and patient infographic/animation;
- A measured improvement in radiographers' knowledge identifying medical implants on radiographs;
- Fewer unexpected implants being introduced to MRI, improving patient safety;
- The completion of all proposed project work packages, per the research timetable;
- The completion of the PhD program and this award.

Risks to delivery:

- Delay in ethics. The research timetable allocates extra time for approvals (9 months)
- Failure to recruit optimal sample sizes:
  Package 2) If participant numbers are insufficient, I'll seek input from the steering group and make targeted feedback requests;

  Package 3) If recruitment appears slow, I will expand the recruitment strategy to teaching hospitals;

  Package 4) The framework states a minimum cohort of six[43]. Here, the infographic will require more extensive field testing e.g., the Cancer Patients' Voice platform and informal feedback through my clinical work.

- eLfH dependence: If all the program modules cannot be published on schedule, a PowerPoint version can be uploaded to CMRAD. Data collection and analysis will be unaffected.

- Completion of the PhD/award: mitigated with regular supervisor meetings (fortnightly) and meeting PhD milestones.

7. Patient & Public Involvement

Please describe how patients/service users, carers and the public have been involved in developing this proposal

Cancer Patients' Voice platform

(https://patients-voice.cancerbrc.org/the-pre-mri-screening-process-what-are-your-thoughts)

Patient feedback revealed common concerns about the MRI screening process, including:

- Insufficient pre-scan information, particularly about the screening process;
- Misunderstandings about the necessity of declaring medically implanted devices, even those implanted since childhood;
- Current pre-scan information is often overlooked/not read by patients;
- Pictures and animations could support MRI screening.

PPI Group

The group met four times and they influenced the following:

Research plan:

- Incorporating feedback from the Cancer Patients' Voice platform, the PPI group contributed to the development of Package 4, the co-creation of an infographic educating patients regarding MRI screening. They also advised on including an animation accompaniment (with subtitles), to better assist some patient groups.

Research design:

- They reviewed and agreed on the timelines for each package;
- The PPI group reviewed the study design for Package 4 and endorsed the recommendations. They suggested recording and summarising meetings for those with memory issues/mental fatigue. Additionally, they proposed limiting online meetings to 2 hours if conducted virtually;
- The meetings should be outside of a hospital and easily accessible by public transport;
- Recruitment should be from different local Trusts and community spaces to increase diversity;
- They advised forming two PPI groups for the proposed project, as it was considered too much for one group to advise on the research and co-create the infographic/animation. However, the project PPI group should be invited to co-facilitate the co-creation meetings. Due to the desired diversity of the co-creation group, a co-facilitator will help ensure all co-creation members are fully supported (e.g., additional verbal explanations and writing assistance) and that their opinions are equally represented;
- Dissemination of the infographic via patient forums, social media, and charity affiliations;
- Review and editing the plain English summary

Professional steering group

This group met once during the PCAF. Their main influence was the promotion and dissemination of the eLearning and infographic interventions. In addition to the UK's radiography professional body, promotion through UK-based MRI organisations (the British Association of MRI Radiographers (BAMRR), MRI Safety Matters, Institute of Medical Physics (IPEM) and 'Everything MRI') or UK radiography-focused podcasts (Rad_Chat, Imagin'That) were suggested.

Please describe the ways in which patients/service users, carers and the public will be actively involved in the proposed research, including any training and support provided

PPI members will receive reimbursement per NIHR payment guidance[51]. The host organisation will provide members with introductory training, including understanding PPI and collaborative strategies. Meetings will be audio recorded and reported using GRIPP2-SF/GRIPP2-LF for PPI and co-creation activities in any published literature[52].

Project PPI

I'll seek up to eight members. We will meet four times in year one to set up the study and organise recruitment materials, then thrice-yearly afterwards. Involvement will include:

- Providing the patient/public perspective on my study and how I can improve/develop it;
- Designing/reviewing all consents, advertisements and participant information for radiographer, co-creation and focus group recruitment;
- Designing/reviewing pre-co-creation focus group questions, ensuring appropriateness and understandability;
- Designing the co-creation project;
- Co-facilitating co-creation meetings;
- Review the final infographic and animation.

Training: As this group will review multiple elements, I will offer 'Training for reviewing' (https://www.learningforinvolvement.org.uk/an-interactive-course-for-new-and-experienced-patient-public-reviewers-of-health-and-social-care-research/). Additionally, as members will be invited to co-facilitate the co-creation meetings, training in facilitation skills will be offered. Some co-creation members may require more assistance/attention to ensure their views are heard and expressed to the group.

Co-creation group

This group will meet up to ten times in-person while developing the infographic. Involvement will include:

- Creating the infographic and animation;
- Collaboratively publish an article on the co-creation process.

Training: For those interested, I will direct group members to the NIHR's co-production webpages, (https://www.learningforinvolvement.org.uk/topic/co-production/). Additionally, I will offer Co-creation training delivered by Arts and Homelessness International. The training can be tailored to our project and the group's needs, especially important considering our group's anticipated diversity.

Both PPI groups will assist disseminating the infographic through charities, community spaces, social media and patient forums.

Cancer Patients' Voice

I will post polls, surveys, and ideas on my page throughout the award to gather informal patient feedback.

Professional steering group

We (managers, educators, safety experts) will meet online at least five times. Involvement will include:

- Discussing eLearning pilot results to inform next steps;
- Advising on the content and NHS integration of the infographic, ensuring all Trusts can incorporate it into their documentation;
- Designing/reviewing eLearning evaluation questions and recruitment assistance;
- Promoting the eLearning training and infographic/animation to the MRI community.

If it is considered not appropriate and meaningful to actively involve patients/service users, carers and the public in your proposed research, please justify why
N/A

8. Training & Development and Research Support

Proposed training and development programme

This program has been developed using the Vitae researcher development framework[53]

PhD program

The Institute of Cancer Research (ICR) has a long-standing clinical imaging research programme, including MRI, and provides training in the skills necessary to excel in research. These range from interpersonal skills and leadership to performing scientific research and project/resource management. There is an extensive catalogue to supplement my learning, some of which are included below. Additionally, I can undertake modules from Perspectives in Oncology. These provide a basic grounding in cancer science including medical physics, structural biology, and cancer treatment, complementing my clinical career.

Research methods

I completed an introductory course in qualitative research methods and analysis, including content analysis. However, further learning to report qualitative research will assist this project:

- Reporting qualitative data, NatCen.

Additionally, as I have no experience in mixed-methods research, I need to complete the following:

- Mixed-methods in health research, University of Oxford.

I lack experience in evaluating complex interventions and statistics. I'll complete the following to help with intervention evaluation:

- Introduction to Statistics and Critical Appraisal/Further Statistics for Health Science Researchers, University of Sheffield;
- Applied Principles and Practice of Behaviour Change, University College London;
- Developing and Evaluating Complex Interventions, NCRM;
- Intervention design and evaluation, University of Sheffield.

I'll take the following to learn how to visualise data effectively and design infographics, promoting more effective communication between the co-creation and professional graphics teams:

- Design and make Infographics, Coursera.

This project requires both co-design and co-creation work. Therefore, I will complete the following:

- Experience-based co-design and Facilitation skills course, The Point of Care Foundation;
- Co-design for All: Doing co-design in practice, Coursera.

Completing the experience-based co-design course allows participation in the Point of Care Foundation's community of practice, an online space to reflect, share ideas and learn from others working in co-design.

I'll attend a Participatory Action Research (PAR) course tailored to those with basic knowledge of qualitative research methods. It covers ethical considerations, establishing rapport with participants, and involving communities in PAR projects:

- Introduction to PAR, SRA.

To enhance my rapport with patients and research participants, I'll attend the following to complement courses completed during the PCAF:

- The Inclusive Communication Course, Course Beetle
- Dealing with difficult situations through forum theatre, NIHR.

I have completed a scoping review, but to perform a systematic review, I need to learn additional skills e.g., quality assessments and meta-analysis:

- Conducting a Systematic Review: A Practical Guide, Cardiff University.

To assist the dissemination and wider impact of my research, I will complete:

- Creating prize-winning scientific posters;
- Presenting research at a conference, ICR.

Complementary courses

I will complete these ICR courses in year one to ensure a strong start to the project:

- Project management for PhDs;
- Research integrity;
- Managing your research project;
- Maintaining momentum during a research project;
- Academic writing Part 1&2.

I will then complete the following to build on this foundation knowledge and strengthen my understanding of governance, ethics and project management:

- Project Management and Research Governance short course, University of Birmingham.

Leadership and Education

I will complete these courses to develop my skills as a future research leader and educator, specifically the development of learning resources, understanding educational theories, the appropriate use of different media and assessing educational impact:

- Managing successful blended teams/Exploring the imposter syndrome for clinical researchers, ICR;
- Emerging clinical leaders programme, The King's Fund;
- Educational media and technologies, Cardiff University;
- Assessing learning, Cardiff University.

Additionally, The Royal Marsden (RMH) is partnered with the Imperial College Academic Health Science Centre, including the Clinical Academic Training Office, where I can access free training, workshops and events to support my learning and networking throughout the award.

Continuing Professional Development

To maintain and develop my clinical and MRI safety-specific knowledge I will attend the following:

- The MRI Safety Officer course teaches MRI staff how to mitigate risks when scanning patients with unknown medical implants and foreign bodies in situ.
- MRI safety and innovation conference, brings together MRI safety experts from around the world to discuss the latest advancements, issues, and innovations in MRI safety.
- UK Imaging and Oncology (UKIO) is an annual, multidisciplinary conference covering all aspects of diagnostic imaging and radiation oncology, radiology informatics, and service delivery. Attendance will be to enhance my clinical knowledge, participate in research hubs (especially in year two where I hope to recruit radiographers for my intervention evaluation), and present my work, if appropriate.
- The Introduction to Artificial Intelligence for radiographers course will introduce how AI is applied in radiography contexts. It explores its strengths/weaknesses, and its impact on practice, workflows and the patient pathway. Completion will aid in planning my proposed post-doctoral work, City University.

Clinical skills

My primary clinical role will remain at RMH, a specialist oncology and tertiary referral centre, that exposes me to patients with complex medical histories and MRI screening challenges. With support from managers and supervisors within the Trust, I'll position myself as an MRI screening specialist, tailoring teaching and training to colleagues and the wider MRI team. Additionally, RMH is building capacity for clinical academic radiographers and applied health research. With guidance and support from my academic/practice supervisors and managers, I aim to transition into a clinical academic role upon completing the award.

I'll continue participating in radiology's journal clubs, and Southwest London Imaging Academy training. I'll participate in RMH's monthly research and audit forums and attend the Trust's annual research day to foster and maintain research networks within my organisation.

To complement my experience, I'll organise secondments to settings that face screening challenges, provide scanning of complex implants and areas where I have no clinical experience:

- Mobile MRI/Community Diagnostic Centres where infrastructure for viewing medical records may be limited, and support from radiologists/medical physicists may be reduced;
- A specialist cardiac centre, where I can familiarise myself with cardiac MRI, one of the UK's largest expansions of all imaging modalities, particularly for imaging patients with non-conditional (untested in MRI) pacemakers;
- 7Tesla (Ultra-high field strength) MRI where independent safety testing of implants/devices is crucial due to the higher field strength not typically encountered in clinical practice.

Primary Doctoral Supervisor

PRIMARY DOCTORAL SUPERVISOR 1

Name	Professor Helen McNair

Institution	Professor of Translational Therapeutic Radiographer and Lead Clinical Academic Radiographer

Position	Royal Marsden Hospital and Institute for Cancer Research

Description of support provided

Professor McNair is fully committed to supervising Hannah to complete a robust PhD. She has >100 publications and is a frequent invited speaker, not only regarding her research areas but also in developing clinical academic roles. She has supervised 14 MSc students, 2 MD's and 3 PhD's. She is currently supervising one PhD, one pre-application NIHR award and one professional doctorate project. She commits to meeting with Hannah once a fortnight.

Previously a holder of a HEEE/NIHR senior clinical lectureship, currently mentoring two NIHR doctoral candidates and PI of a NIHR incubator for diagnostic and therapeutic radiographers, she will be able to apply her extensive experience in creating clinical academic pathways to advise Hannah.

Her research interests include developing training, assessing competency, and implementing new workflows. As a therapeutic radiographer working on a MR-Linac she is acutely aware of the MRI safety screening issues. Via training in her NIHR grant she has improved her skills in qualitative research to enable her to co-supervise all aspects of the research. She has supervised Hannah's predoctoral work and looks forward to continuing to be part of Hannah's journey in becoming a clinical academic and effective researcher.

As a ICR team leader, Helen will use her links to ensure Hannah has access to relevant facilities and training. She works closely with the clinical academic training office at Imperial College , liaises with the Biomedical Research Centre, a partnership between RM and ICR and works alongside Professor Susanne Cruickshank, Applied Health Research lead at RM.

Additional Academic Supervisor(s)

ADDITIONAL ACADEMIC SUPERVISOR 1

Name	Dr Christina Malamateniou

Institution	City University of London

Position	Reader/Associate Professor in Radiography

Description of support provided
Dr Christina Malamateniou is an Associate Professor in radiography at City University of London and the research lead of the Department of Radiography. She is an Adjunct Professor at UCC in Ireland, Associate Editor for JMIRS and Board member of EuSoMII. She has received multiple awards and prizes for her research in MRI and AI. Her academic career spans 21 years, she has more than 85 published papers in peer-reviewed journals, h-index 22, grant funding of £3.25mil from different funders as PI and Co-I, and has co-authored 6 standards/guidelines on AI, MRI safety, and paediatric MRI. She is the current chair of research for the EFRS (2023-2025). She has led the work on the development of the MRSO/MRSE and MRSD MRI safety standards for the EFRS in Europe (MRSO Role Descriptor: an EQF benchmarking document (efrs.eu)) and is publishing actively in this field. She has seen 2 PhD students to completion, one of which is an NIHR doctoral fellow, examined as external examiner 8 PhDs and currently supervises another 3 PhD students, all radiographers. There are 3 research strands in her research outputs and departmental strategy: A) technological innovation in medical imaging, B) patient-centred care in medical imaging and C) patient safety and patient care. This project aligns fully with the last one, but also touches the other two areas of expertise.

ADDITIONAL ACADEMIC SUPERVISOR 2

Name	Professor Susanne Cruickshank

Institution	The Royal Marsden NHS Foundation Trust

Position	Strategic Lead for Applied Health Research, Health Services Research

Description of support provided
Professor Susanne Cruickshank is the Strategic Lead for Applied Health Research at the Royal Marsden, a cancer nurse by profession and with senior leadership experience across research, education, and practice. She has 6 PhD completions (n=4, 1st Supervisor; n=2 2nd Supervisor) and over 20 MRES/MSc (n=20 as 1st supervisor). Within her role she leads clinical academic research capacity and capability development among Nurses, Allied Health Professionals and Pharmacists while working closely with the NIHR BRC/ICR to share resources. She leads the Health Services Research Group at RM, whose portfolio of funded research from NIHR and others focuses on improving the lives and outcomes of people affected by cancer. The work is organised around three distinct areas: A) Systematic reviews of the evidence/theory to inform care decisions and future research; B) Using qualitative and mixed methods to improve the experiences, unmet needs, and outcomes of cancer patients; C) Developing, testing, and implementing complex physical, psycho-social, and emotional supportive care interventions in healthcare settings. It is this depth and breadth of experience that will complement my supervisory team.

Clinical Supervisor(s)

PRACTICE SUPERVISOR 1

Name	Dr Ian Simcock

Institution	Great Ormond Street Hospital for Children NHS Foundation Trust

Position	Clinical academic radiographer

Description of support provided
Dr Ian Simcock is the lead Clinical-Academic Radiographer for GOSH Radiology and completed his NIHR funded PhD in 2021. He developed non-invasive micro-CT for human fetal post- mortem imaging following miscarriage, winning several awards including the NIHR Biomedical Research Centre Excellence in Impact Award 2023. He published this work in Nature and developed a public dissemination animation video with multiple charities to explain post-mortem choices to parents. He has recently been awarded a £200k NIHR RfPB award to continue working in this area and is named on £1million grants. Dr Simcock has completed further leadership and mentorship training including the HEE LSBU post-doctoral bridging scheme, Kings Fund - Emerging Clinical Leaders and the GOSH - Developing Leaders Programme as well as the Society of Radiographers mentorship scheme training. He has been elected as Chair of the Society of Radiographers Research Group, sits on the Editorial Board of Radiography Journal, is an external advisor to the NIHR Mentoring programme at UWE, and is on the Junior Faculty of the GOSH Biomedical Research Centre. He has successfully mentored Hannah and multiple other radiographers through their PCAF awards, providing guidance and support to transition through their early research career. His knowledge and awareness of clinical-academic careers will help Hannah progress, to grow her profile and influence within the national and international radiographic and wider research community and help her develop into a future clinical-academic leader.

PRACTICE SUPERVISOR 2

Name	Dr Marco Borri

Institution	King's College Hospital NHS Foundation Trust

Position	Lead clinical scientist

Description of support provided
I have known Hannah for more than 6 years, and I have acted as a supervisor and mentor since her time at the Neuroradiology Department at King's College Hospital NHS Foundation Trust. I have collaborated with Hannah on several projects related to MRI safety, and I supported Hannah in presenting them at national (IPEM) and international (ISMRM) leading conferences on MRI safety and to publish a scientific article in Radiography. This also led to our ongoing collaboration with eLearning for Healthcare (NHS England). I have been a clinical and academic supervisor, and mentor, for 12 MSc students, including Hannah. As a Clinical Scientist and former NIHR doctoral fellow, I am in a strong position to advise and support Hannah on the wider aspects of both her professional development and her fellowship project. An integral part of my current role is to empower radiographers to broaden their horizons and develop their clinical role, by training them and providing them with professional and scientific support. As a certified MR Safety Expert and member of national and international boards on MRI Safety, I am also best placed to advise Hannah on the wider reach of her project, making sure that her fellowship and career fit into the broader conversation on MRI Safety. I have discussed these aspects with Hannah in preparation for her application, and I plan to meet with her for an hour on a bi-monthly

basis during the fellowship to support her in her clinical and professional development.

PRACTICE SUPERVISOR 3

Name	Erica Scurr DCR (R) MSc MRSO (MBMS TM)

Institution	The Royal Marsden NHS Foundation Trust

Position	Cross-Site Lead MRI Superintendent Radiographer

Description of support provided

I am the MRI lead radiographer responsible for clinical MRI and research services at the Royal Marsden. I am a clinical specialist having worked in MRI for 32 years and specialising in oncology for over 20 years. I am a respected professional, leader and educator in this field.

Since Hannah joined the Trust, I have encouraged her to develop her work as I believe it has a pivotal role in the education of patients and staff, highlighting the importance of safe practices in MRI. With the rapid growth in MRI services, it is important to implement accessible learning to support our workforce in all aspects of MRI safety. Implanted devices present a particular challenge, especially in compromised and vulnerable patients. Recognising implants before MRI is critical to positive patient outcomes. This has been known since MRI's clinical inception, but Hannah has had the vision and drive to formalise implant identification which will be welcomed by the MRI community and could have national and global reach.

I am passionate about patient safety and the benefit that the e-learning module will have in supporting staff to ensure safe, high-quality patient-centred care. The infographic will inform patients and empower them to work with radiographic professionals, ensuring safety during their imaging pathway.

In my clinical lead role, I will be available to work with Hannah to ensure that she receives the support required to meet the aims of this valuable project and provide mentorship as she continues to develop her leadership and academic career.

Collaborations

The Society and College of Radiographers (SCoR) and eLearning for Healthcare (eLfH, NHS England).

My eLearning training intervention will be held in the Clinical Imaging section of the eLfH platform (https://www.e-lfh.org.uk/programmes/clinical-imaging/), where I am listed as an author.

eLfH works in partnership with the NHS to support patient care by providing free eLearning sessions to educate and train the health and social care workforce. eLfH programs are increasingly utilised on staff electronic records for statutory and mandatory training, enhancing familiarity with the system. With 24/7 access, this platform will allow clinical staff to view and test their learning at a pace and time that suits them. As part of that, Clinical Imaging is designed to support imaging staff of all grades and specialist modalities by providing a range of educational content including image interpretation, patient care, imaging patients with specific needs and clinical governance. The Clinical Imaging eLearning programme is developed and endorsed by the Society and College of Radiographers, in collaboration with NHS England.

I was keen to forge this collaboration with eLfH as it will allow every radiographer in the NHS to have free access to this training program. Additionally, it can be accessed by any healthcare professional in the NHS, or through OpenAthens, with an interest in implant identification such as nurses, medical students, radiology trainees and other Allied Health Professional groups.

I am working with a small team to build and implement the programme on the platform:

- Dorothy Keane- Clinical lead of the Clinical Imaging project for the Society and College of Radiographers;
- Dave Beardmore- Senior project manager for NHSE eLearning for healthcare, specialising in Technology Enhanced Learning (TEL) education;
- Alan Campbell- Learning designer for NHSE eLearning for Healthcare;
- Marco Borri- MRI clinical scientist and MRI safety expert

Additionally, eLfH content is accessible to organisations outside of the NHS for a fee (https://www.eintegrity.org/healthcare-course/clinical-imaging/). With the right advertising, this will allow the training program to reach an international audience, the promotion of advanced screening methods globally and increased revenue for the NHS.

NHS Informatics Merseyside

NHS Informatics Merseyside is an NHS digital technology organisation that ensures technology and innovation are used to enable better health and care (https://www.imerseyside.nhs.uk/). They will be collaborating with the co-creation team and myself to create our patient infographic and animation.

Their illustration services will produce two versions of the infographic, to support print and electronic distribution. Their designers are Adobe Certified Associates in graphic design and illustration and will ensure our co-created design will engage with the public. Their motion graphic and interactive services will produce our animation. The illustrated animation will be created in alignment with the campaign content and styling to reinforce the key messages contained within the infographic to engage patients using an alternative format. The design of both products will be produced in accordance with the NHS Identity Guidelines and accessibility standards.

Following NHS Identity Guidelines is essential to embed the infographic into local communications, which will likely require the infographic to be branded by the organisations that choose to use it. This includes features such as fonts, service branding and tone of voice (e.g., concise, professional and accessible language). Additionally, by creating an online animation, it must be available for use on a website or mobile device by as many people as possible (e.g., people with motor difficulties, cognitive impairments/learning disabilities or the hearing impaired).

The development of the infographic will allow adjustments and changes to be made and the animation will be developed through a storyboard, before finalising the products.

Host Organisation Support Statement

The following statement of support should be composed jointly by the Heads of Department / Senior Managers of the employing host organisation and the partner organisation.
The Heads of Department / Senior Managers of these organisations will be able to add the statement of support to the form once they have confirmed participation. Invitations to participate are sent by the applicant via the 'Participants and Signatories' section of the form.

Statement of support

We are delighted to support Hannah's application for an NIHR Doctoral Clinical and Practitioner Academic Fellowship (DCAF). Hannah is currently a Senior Magnetic Resonance Imaging Radiographer at The Royal Marsden NHS Foundation Trust (RM) and the fellowship will be hosted in the RM radiotherapy department in partnership with The Institute of Cancer Research (ICR) in the Division of Radiotherapy and Imaging. Both organisations have a long-standing relationship through the NIHR RM/ICR Biomedical Research Centre (BRC) and are experienced in supporting clinicians such as Hannah to practise during their fellowships, combining research roles with clinical duties. The supervisory team is led by Professor Helen McNair, a highly experienced supervisor with 5 completions of MD/PhD and over 14 MSc students. Hannah's supervisors and wider clinical team have proudly supported Hannah's development during her NIHR PCAF and are committed to develop aspiring clinical academic radiographers during, and after doctoral studies.

The Radiography Department:

Radiography at RM is a recognised centre of clinical excellence combined with successful academic research. Professor Helen McNair, Hannah's primary supervisor is herself a HEE/NIHR Senior Clinical Lecturer (2019-2024), our Trust Lead Research Radiographer. She is Chair of the NIHR Diagnostic and therapeutic Incubator, a collaboration with Imperial College Academic Health Science Centre. Professor McNair has established a vibrant community of research-interested and -active radiographers which strives to increase research capacity and capability across all radiographers. She will be able to play a key role in Hannah's development as a clinical academic.

Hannah has been motivated by this positive and supportive research environment to find solutions to a clinical issue she is particularly interested in, MRI safety. Hannah's specific clinical expertise in this area, her previous work in building networks nationally and internationally, and her work on the PCAF makes her an excellent candidate for an NIHR DCAF. We have no doubt that she will use this fellowship to improve the safe care for all patients receiving MRIs. The supervisory team includes Dr Christina Malamateniou, Associate Professor in radiography at City University of London and research lead of the department of radiography. She brings extensive expertise in Hannahs area of interest, MRI safety but also experience of supervision, using mixed methods and AI. Hannah will have access to the innovative training Christina offers in AI and other emerging areas in radiography at City.

The NIHR RM/ICR Biomedical Research Centre (BRC):

The NIHR RM/ICR BRC is the only BRC specifically focused on cancer. The research strategy seeks to inspire the future generation of researchers and to ensure equality, diversity and inclusion is visible to all. Hannah's work aligns with the Imaging and Data Science theme of our BRC as MRI is increasingly used in the cancer diagnosis, treatment, and staging pathway. The BRC's main priority is to support translational research, and the work Hannah proposes aligns very strongly with these intentions.

The Institute of Cancer Research

Hannah will be part of the Division of Radiotherapy and Imaging of the ICR, led by Professor Kevin Harrington. The division includes the prestigious Centre for Cancer Imaging and has long-standing clinical imaging research programmes in computed tomography (CT) and magnetic resonance imaging (MRI). The ICR holds the European Commission HR Excellence in Research Award and is a signatory of the revised Concordat to support the career development of researchers and Concordat to support research integrity. ICR has a strong academic supportive environment and research community, over the 2021/22 academic year, there were 264 people registered on research degrees with the ICR including 41 new fully funded PhD students. Of ICR non-clinical PhD students graduating between 2010 and 2019, 91% went directly into research roles.

Training and development for Hannah include courses related to research skills, statistics and scientific computing and careers and leadership activities that are specifically tailored to researchers at Hannah's stage, but this will be complimented with links the supervisors have to professional specific groups and opportunities.

The Institute of Cancer research (ICR), and the RM place equality, diversity, and inclusion (EDI) at the forefront of their joint research strategy, hosting regular events to promote its importance and monitor actions. Both organisations attract staff and students from across the world from completely different backgrounds with a variety of religious, sexual and gender identities. We are proud of this incredible diversity.

The Royal Marsden NHS Foundation Trust:

The RM prides itself on its track record for developing clinical academic careers for Nurses, Allied Health Professionals and Pharmacists (NAHPPs) and is fully supportive of Hannah's application. She is already part of our community of researchers pursuing a clinical academic pathway including staff on internships [n= 1 HEE/NIHR programme; n=4 internally], pre-doctoral fellowship [n=9 internally funded; n=4 NIHR PCAF] and doctoral programmes [n=11, 2 on NIHR DCAF] and NIHR SCPRA awards (n=2). All staff on these programmes are expected to participate in a monthly research meeting. Hannah gets an opportunity to share her work with the group and learn from staff who are further along the pathway i.e. staff doing an NIHR doctoral fellowship or who can share their experiences of patient and public engagement as early-career researchers. In addition, we have 20 NAHPPs who are post-doctoral and act as role models throughout the Trust.

Professor Susanne Cruickshank, the Strategic Lead for Applied Health Research, is one of 15 Research Leads in RM. She has overall responsibility for the clinical academic programmes for NAHPPs within the Trust. As an experienced researcher, cancer expert and educator she provides leadership and guidance so staff can flourish as researchers and will continue to act as one of Hannah's supervisors. She has 6 PhD and over 20 MRES/MSc completions across different disciplines and brings a breadth of research methodological experience.

Hannah would continue to be supported beyond the completion of this doctoral fellowship to enable her to develop her future clinical academic career by our organisations. Overall, this collaborative support from The Royal Marsden, ICR and her supervisory team represents an unparalleled training opportunity for Hannah and foundation for a leadership role in research.

9. Detailed Budget

The finance section should provide a breakdown of costs associated with undertaking the research as described in the proposal.

1

Staff type	Lead Applicant

Name	Hannah Barnsley

Employing organisation	The Royal Marsden NHS Foundation Trust

Grade	7

Starting point on scale	8

Increment Date	01/05/2024

Justification of Costs
This is a breakdown of my predicted full salary costs, including High Cost Area Supplement (HCAS).

Details of Posts and Salaries Costs

	Total (£)
Salary (£)	50056
Geographical Weighting (£)	7746
Other allowances (£)	0
Superannuation (£)	8312
Nat. Insurance (£)	6721
Current Annual Costs (£)	**72835**

Direct Costs
These are the costs that are specific to the research, which will be charged as the amount actually spent and can be supported by an audit record.

Staff Type: (Lead Applicant)-Miss Hannah Barnsley
Type of cost: NHS
% Full-time on this research: 100.0
Total months on this research: 36.0
Justification of Cost: This is the salary costs of the lead applicant so I can carry out the research and training outlined in this proposal.

	Year 1	Year 2	Year 3	Total Cost (£)
Basic Salary	50056	50056	50056	150168
National Insurance	6721	6721	6721	20163
Superannuation	8312	8312	8312	24936
Geographical weighting / Other allowances (if applicable)	7746	7746	7746	23238

Staff Type: Other
Other Type: Shared Staff
Name: **Statistician**
Type of cost: NHS
% Full-time on this research: 5.0
Total months on this research: 3.0
Justification of Cost: I will perform all statistical analysis for my study but the statistician will be available to answer any queries that arise during the fellowship.

	Year 1	Year 2	Year 3	Total Cost (£)
Basic Salary	3165	3165	3165	9495

Totals	Applicant	Support Staff	Shared Staff	Total Cost (£)
Total HEI	0	0	0	0
Total NHS	218505	0	9495	228000
Total Commercial	0	0	0	0
Total Other	0	0	0	0

Travel, Subsistence and Conference Fees

Type of cost: NHS
Description: Conference Fees
Conference: True
Item Description: UKIO and MRI safety update
Justification of Cost: Both the UK Imaging and Oncology (UKIO) conference and MRI safety update directly relate to my continuing professional development as a cancer specialist practitioner and MRI radiographer. Throughout the DCAF they will allow me to disseminate the different stages of my work to the relevant audiences. UKIO covers all aspects of diagnostic imaging and radiation oncology, radiology informatics, and service delivery. Attendance will be to enhance my clinical knowledge, participate in research hubs (especially in year two where I hope to recruit radiographers for my intervention evaluation), and present my work, if appropriate. The MRI safety update brings together MRI safety experts from around the world to discuss the latest advancements, issues, and innovations in MRI safety. Costs are based on 2024 rates. These costs are reflective of being an attendee. However, if presenting at the MRI safety matters conference (as I am in November), all fees will be covered by the organisers.

	Year 1	Year 2	Year 3	Total Cost (£)
	360	360	360	1080

Type of cost: NHS
Description: Journey Costs
Conference: True
Item Description: UKIO and MRI safety update train travel
Justification of Cost: UKIO is held in Liverpool every year and the MRI safety update is held in Edinburgh. Return rail fare for both will be purchased in advance to try and get a cheaper fare and with a personal Network Rail card where possible (restrictions apply).

	Year 1	Year 2	Year 3	Total Cost (£)
	220	220	220	660

Type of cost: NHS
Description: Subsistence
Conference: True
Item Description: UKIO and MRI safety update accommodation
Justification of Cost: These costs cover accommodation at each conference (UKIO 4 nights and MRI safety update 2 nights).

	Year 1	Year 2	Year 3	Total Cost (£)
	420	420	420	1260

	Year 1	Year 2	Year 3	Total Cost (£)
Travel, Subsistence and Conference Fees Total	1000	1000	1000	3000

Equipment

Type of cost: NHS
VAT relief to be applied: False
Description: Digital Voice Recorder
Price excluding VAT £: 40
Expected life time: 36
Justification of Cost: This is a digital voice recorder for use during the co-creation meetings. There is expected to be at least 14 people present at each meeting in a fairly open space. In order to accurately record each meeting, a device specifically designed for this function is needed. It can also be uploaded to a computer and is digitally time-stamped.

	Year 1 (exc VAT)	Year 2 (exc VAT)	Year 3 (exc VAT)	Total Cost (£) (exc VAT)
	40	0	0	40

Type of cost: NHS
VAT relief to be applied: False
Description: Laptop
Price excluding VAT £: 950
Expected life time: 36
Justification of Cost: This is required to complete this study, including writing of study. The majority of training is conducted online as well as PPI activities and the systematic review.

	Year 1 (exc VAT)	Year 2 (exc VAT)	Year 3 (exc VAT)	Total Cost (£) (exc VAT)
	950	0	0	950

	Year 1	Year 2	Year 3	Total Cost (£)
Equipment Total	990	0	0	990

Consumables

Description: Consumables for in-person co-creation meetings
Type of cost: NHS
Please describe fully: Consumables for in-person co-creation meetings
Justification of Cost: The co-creation meeting will require the generation of ideas from the members. Guidance recommends the use of paper charts/sticky notes. This cost covers x2 rolls of recyclable brown paper, which can be affixed to the walls and written on (£15 each from Amazon), an assortment of sticky notes for people to write on (£35 from Amazon), a pack of pens (£20 from Amazon) and masking adhesive table (£4 from Amazon).

	Year 1	Year 2	Year 3	Total Cost (£)

	89	0	0	89

Description: Recruitment materials
Type of cost: NHS
Please describe fully: Recruitment materials
Justification of Cost: This covers the cost of paper/printing of recruitment materials (e.g., posters/flyers), participant information sheets/consent forms for PPI work and postage. This is the cost over the 3-year duration.

	Year 1	Year 2	Year 3	Total Cost (£)
	300	0	0	300

	Year 1	Year 2	Year 3	Total Cost (£)
Consumables Total	389	0	0	389

Patient and Public Involvement

Description: PPI host organisation induction training
Type of cost: NHS
Please describe fully: PPI host organisation induction training
Justification of Cost: This covers a 4-hour induction training session for all 20 PPI members (PPI group and co-creation group, at £75/person), delivered by the Royal Marsden PPI department. These sessions are currently held online so a further contribution of £5/person is also included to cover internet costs. This also covers the costs for caring responsibilities while an individual completes the induction training. Rates are £25/hour (based on Age UK's homecare costs https://www.ageuk.org.uk/information-advice/care/paying-for-care/paying-for-homecare/) for 20 people.

	Year 1	Year 2	Year 3	Total Cost (£)
	3600	0	0	3600

Description: Project PPI attendance costs
Type of cost: NHS
Please describe fully: Project PPI attendance costs
Justification of Cost: This covers the time of the project PPI team of 8 individuals at £50 per 2-hour session. I expect to meet with this group 10 times over the course of this project.

	Year 1	Year 2	Year 3	Total Cost (£)
	1600	1200	1200	4000

Description: Project PPI travel + meeting location costs
Type of cost: NHS
Please describe fully: Project PPI travel + meeting location costs
Justification of Cost: This group will be recruited through the Royal Marsden PPI department. Therefore, they could reside in the greater London region or surrounding counties. It is expected that these meetings will happen virtually with an additional payment of £5 for internet usage being offered on top of attendance costs. However, if the group want to meet in person, travel costs (up to £100 for national travel) will be offered to attend the meeting location at Camberwell Library (£64.10/hour, see co-creation meeting location description) and subsistence costs (£5/person per meeting based on NIHR recommendations. In-person meetings have been costed. https://www.nihr.ac.uk/documents/payment-guidance-for-researchers-and-professionals/27392)

	Year 1	Year 2	Year 3	Total Cost (£)
	3873	2905	2905	9683

Description: Project PPI carers costs
Type of cost: NHS
Please describe fully: Project PPI carers costs
Justification of Cost: This covers the costs for caring responsibilities for 8 PPI members while an individual attends the PPI meetings. Rates are £25/hour (Age UK's estimated hourly rate https://www.ageuk.org.uk/information-advice/care/paying-for-care/paying-for-homecare) for 8 people

	Year 1	Year 2	Year 3	Total Cost (£)

	1600	1200	1200	4000

Description: Co-creation PPI attendance
Type of cost: NHS
Please describe fully: Co-creation PPI attendance
Justification of Cost: This is the total amount for 12 co-creators' time, for 10 3-hour sessions (25/hr) in accordance to the research timetable. The time will be spent co-creating a patient information infographic resource with an accompanying animation.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	9000	0	9000

Description: Co-creation PPI travel costs
Type of cost: NHS
Please describe fully: Co-creation PPI travel costs
Justification of Cost: It is expected this group will be local to Southeast London so only local travel costs will be incurred of £10 per session.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	1200	0	1200

Description: Co-creation PPI caring costs
Type of cost: NHS
Please describe fully: Co-creation PPI caring costs
Justification of Cost: This covers caring costs for each member of the co-creation team for the duration of the meetings at £25/hour (Age UK's homecare rates https://www.ageuk.org.uk/information-advice/care/paying-for-care/paying-for-homecare/) for 12 people.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	9000	0	9000

Description: Co-creation meeting location costs
Type of cost: NHS
Please describe fully: Co-creation meeting location costs
Justification of Cost: Camberwell library is central to Southwark, in a transport hub and in close proximity to King's College Hospital. It has WiFi, an integrated screen and projector and access to a kitchen with accessories. Additionally, it supports wheelchair access to both floors, disabled toilets, access to Dolphin Supernova software (allows screen magnification and audio assistance for visually impaired) and allows assistant dogs. It costs £64.10/hour. https://www.southwark.gov.uk/leisure-facility-hire-and-bookings/3/camberwell-library-meeting-room-3

	Year 1	Year 2	Year 3	Total Cost (£)
	0	1924	0	1924

Description: Co-creation PPI subsistence costs
Type of cost: NHS
Please describe fully: Co-creation PPI subsistence costs
Justification of Cost: This will be used to provide tea/coffee/refreshments for the co-creation meetings based on £5/person, based on NIHR recommendations (https://www.nihr.ac.uk/documents/payment-guidance-for-researchers-and-professionals/27392)

	Year 1	Year 2	Year 3	Total Cost (£)
	0	600	0	600

Description: Co-facilitator training attendance
Type of cost: NHS
Please describe fully: Co-facilitator training attendance
Justification of Cost: Members of the project PPI team will help facilitate the co-creation meetings. This cost will cover the time for 8 people attending a 1-day, in-person training session (£150), subsistence costs (£10 for 10 hours away from home), travel (up to £100 for national travel) and caring costs (25/hour for 1 8-hour day per Age UK recommendations).

	Year 1	Year 2	Year 3	Total Cost (£)

	3680	0	0	3680

Description: Co-facilitator training course
Type of cost: NHS
Please describe fully: Co-facilitator training course
Justification of Cost: This 1-day, in-person course will help improve facilitation skills and confidence to facilitate both online and face-to-face events. It will provide guidance on how to involve a wide range of voices and ensure everyone feels heard, give ideas for enabling people to work together and get the best out of groups and explore issues commonly faced by facilitators. https://www.pointofcarefoundation.org.uk/event/one-day-facilitation-course/

	Year 1	Year 2	Year 3	Total Cost (£)
	3160	0	0	3160

Description: Co-facilitator meeting expenses
Type of cost: NHS
Please describe fully: Co-facilitator meeting expenses
Justification of Cost: This cost covers the attendance of one PPI co-facilitator per co-creation meeting over 10 meetings total. Time (£75 for 1 3-hour meeting), subsistence costs (£5), travel costs (up to £100 for national travel) and caring costs (£25/hour for 3 hours per Age UK recommendations).

	Year 1	Year 2	Year 3	Total Cost (£)
	0	2250	0	2250

Description: Project PPI training for reviewing course
Type of cost: NHS
Please describe fully: Project PPI training for reviewing course
Justification of Cost: This is an free, online course, run by the NIHR, specifically for patient and public reviewers of health and social care research. While this training focuses on reviewing applications and more complex documentation, the skills learnt will be transferable to this project's requirements. It is recommended 1-2 hours for each of the three modules. I have costed for the maximum time of 2 hours per module (£150 per person), internet costs (£5) and caring costs (£25/hour)

	Year 1	Year 2	Year 3	Total Cost (£)
	1525	0	0	1525

Description: Co-creation training for the co-creation team
Type of cost: NHS
Please describe fully: Co-creation training for the co-creation team
Justification of Cost: This cost covers the co-creation team's training regarding the co-creation process. The training will be offered before the first meeting and provided by Arts and Homelessness International (https://artshomelessint.com/). I first heard about this organisation during my PCAF, when I attended a session they ran introducing co-creation. Having decided to use co-creation, I have struggled to find healthcare-related co-creation training for my team (https://artshomelessint.com/what-we-do/training/co-creation/). They can run a workshop (up to 4 hours) and tailor it to our project and the team's needs, especially important given the expected diversity of the group. It can be run online or in person, with the stated costs relating to an in-person meeting. Training session provided by Arts and Homelessness International: £910 (includes a 4-hour workshop and time to prepare the training). Participant costs include their time (£75/person for the session), travel (£10/person), location costs (Camberwell library £64.10/hour) and carers costs (£25/hour) for all 12 members of the team.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	3387	0	3387

Description: PPI co-facilitator for two focus groups
Type of cost: NHS
Please describe fully: PPI co-facilitator for two focus groups
Justification of Cost: This covers the cost for one PPI member to co-facilitate the four online focus groups (patient and radiographer) before and after the co-creation process. It includes their time (£50 for each session max. 3 hours), online costs (£5/meeting) and caring costs (£25/hour)

	Year 1	Year 2	Year 3	Total Cost (£)
	210	210	0	420

	Year 1	Year 2	Year 3	Total Cost (£)
Patient and Public Involvement Total	19248	32876	5305	57429

Other Direct Costs

Description: Collective Minds Radiology platform
Type of cost: NHS
Please describe fully: Collective Minds Radiology platform
Justification of Cost: This study requires radiographer participation in varying geographical locations to view banks of images. Sharing large numbers of anonymised patient images to multiple NHS Trusts using the standard methods of patient image transfer (IEP) are not appropriate. The platform can be accessed from anywhere and will hold the image banks (as well as the qualitative surveys) for the study. The images will only be available to the registered participants of the study through personalised passwords. The site is also fully GDPR compliant. Studies conducted at the host organisation have used CMRAD previously, with success. The quote was for 18 months and has been allocated per the research timetable. https://about.cmrad.com/

	Year 1	Year 2	Year 3	Total Cost (£)
	0	7334	3667	11001

Description: Pre/post co-creation focus group patient and public participation costs
Type of cost: NHS
Please describe fully: Pre/post co-creation focus group patient and public participation costs
Justification of Cost: This covers the time for 10 patient focus group members for two sessions to discuss their priorities for an MRI screening infographic/animation resource and their thoughts on the finalised version and how best it can be implemented, for a maximum of 3 hours at £50. This meeting will be held online, so a cost of £5/person to cover internet costs will be offered. This sum also includes any caring needs for the focus group members at £25 per hour (Age UK https://www.ageuk.org.uk/information-advice/care/paying-for-care/paying-for-homecare/).

	Year 1	Year 2	Year 3	Total Cost (£)
	1300	1300	0	2600

Description: Transcription service
Type of cost: NHS
Please describe fully: Transcription service
Justification of Cost: This is the cost of a subscription service named 1st Class Secretarial Services to transcribe all the focus group and co-creation group meetings. The service is anticipated to be at a Group 5 rate for focus groups with up to 14 people involved @ £2.60 per minute with a total cost of £6240 plus VAT (Based on 40 hours worth of audio – up to 14 speakers per session). 1st Class Secretarial services are fully GDPR compliant and there is a pre-existing confidentiality agreement in place with The Royal Marsden. https://www.1stclass.uk.com/

	Year 1	Year 2	Year 3	Total Cost (£)
	0	7488	0	7488

Description: NHS Informatics Merseyside
Type of cost: NHS
Please describe fully: NHS Informatics Merseyside
Justification of Cost: This is the quote from NHS Informatics Merseyside for the production of the patient infographic (£2410) and accompanying animation (£3615), which will be developed with my co-creation team. The quoted price is for the production of the final infographic. However, as in any design process, there will be tweaks and adjustments to get this to where it needs to be so this is included as part of the price. These costs are exclusive of VAT, which is not charged for payments between NHS organisations. https://www.imerseyside.nhs.uk/

	Year 1	Year 2	Year 3	Total Cost (£)
	0	6025	0	6025

Description: Radiographer recruitment/participation costs
Type of cost: NHS
Please describe fully: Radiographer recruitment/participation costs
Justification of Cost: This cost covers the time for 100 radiographers to complete the training evaluation. This

will be used to backfill time to cover radiographers being released from clinical duty, otherwise the research becomes infeasible. However, the costs listed are only expected if we struggle to recruit radiographers and have to recruit directly from NHS hospitals and open them as sites. This money will only be sought if our other recruitment methods (online, professional body etc) are not attracting sufficient numbers. Therefore, it is not expected that this cost will be required. We have costed vouchers (£25/radiographer) in the application, awarded to radiographer participants on completion of the 6-month post-test and questionnaire. Please see the SoECAT form for the breakdown.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	23500	0	23500

Description: Radiographer recuitment/participation in pre/post co-creation focus groups
Type of cost: NHS
Please describe fully: Radiographer recuitment/participation in pre/post co-creation focus groups
Justification of Cost: This cost covers the time for up to 10 radiographers to participate in up to two focus group sessions regarding the co-created infographic and animation. The costs listed are only expected if we have recruited directly from NHS hospitals and open them as sites. This money will only be sought if our other recruitment methods (e.g., online) are not attracting sufficient numbers. Therefore, it is not expected that this cost will be required. Please see the SoECAT form for the breakdown.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	1645	0	1645

Description: Thank you gesture for radiographer participants
Type of cost: NHS
Please describe fully: Thank you gesture for radiographer participants
Justification of Cost: This cost covers a £25 voucher for each radiographer who participates in the eLearning training evaluation. It will be given as a thank you gesture after the 6-month test/questionnaire has been completed. As we hope most radiographer participants will be recruited through social media or our professional body/conferences, this will be the preferred cost. If radiographers need to be directly recruited through NHS hospitals, this money won't be used and we would seek to use the funds detailed in the SoECAT form.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	2500	0	2500

	Year 1	Year 2	Year 3	Total Cost (£)
Other Direct Costs Total	1300	49792	3667	54759

Training And Development

Type of cost: NHS
Description: Short Courses
Item Description: Reporting qualitative data
Justification of Cost: Delivered by the national Centre for Social Research, this course teaches core skills for writing outputs based on analysis of qualitative data.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	220	220

Type of cost: NHS
Description: Short Courses
Item Description: Mixed methods in health research
Justification of Cost: Delivered by the University of Oxford, this course focuses on ways to combine qualitative and quantitative methods to develop a variety of mixed methods studies and give a deeper understanding of complex healthcare issues. https://www.conted.ox.ac.uk/courses/mixed-methods-in-health-research

	Year 1	Year 2	Year 3	Total Cost (£)
	3175	0	0	3175

Type of cost: NHS
Description: Short Courses

Item Description: Introduction to Statistics and Critical Appraisal
Justification of Cost: Delivered by the University of Sheffield, this course introduces basic concepts and statistical techniques (e.g., hypothesis testing and confidence interval estimation). It will teach simple statistical methods and the principles behind some advanced methods (e.g., regression). This course needs to be completed before continuing onto the advanced statistics course. https://www.sheffield.ac.uk/smph/modules/har6042-introduction-statistics-and-critical-appraisal-online

	Year 1	Year 2	Year 3	Total Cost (£)
	946	0	0	946

Type of cost: NHS
Description: Short Courses
Item Description: Further Statistics for Health Science Researchers
Justification of Cost: The second statistics course delivered by the University of Sheffield teaches fundamental statistical concepts, and simple and advanced statistical methods of multiple regression, survival analysis and generalised linear models. https://www.sheffield.ac.uk/smph/modules/har6045-further-statistics-health-science-researchers

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	946	946

Type of cost: NHS
Description: Short Courses
Item Description: Applied Principles and Practice of Behaviour Change
Justification of Cost: Delivered by University College London, this course is aimed at intervention designers. It will provide a greater understanding of applying behaviour change to interventions and a practical framework for developing and implementing these interventions. https://www.ucl.ac.uk/behaviour-change/intermediate-applied-principles-and-practice-behaviour-change

	Year 1	Year 2	Year 3	Total Cost (£)
	0	3760	0	3760

Type of cost: NHS
Description: Short Courses
Item Description: Developing and Evaluating Complex Interventions
Justification of Cost: This 5-session course is run by the National Centre of Research Methods (MCRM) and the University of Glasgow. Structured around the MRC/NIHR Framework for Developing and Evaluating Complex Interventions, it will introduce the main concepts of developing/identifying and evaluating complex interventions and help me apply them to my own work. https://www.ncrm.ac.uk/training/show.php?article=13285

	Year 1	Year 2	Year 3	Total Cost (£)
	375	0	0	375

Type of cost: NHS
Description: Short Courses
Item Description: Intervention design and evaluation
Justification of Cost: Delivered by the University of Sheffield, this course will look at the evaluation of complex intervention in the 'real world'. It will expand on the previous training on complex interventions by focusing on health and social care-related interventions where the evaluator has limited control over the intervention or context. https://www.sheffield.ac.uk/smph/modules/real-world-evaluation-ten-key-principlesevaluating-complex-health-and-social-interventions-online

	Year 1	Year 2	Year 3	Total Cost (£)
	450	0	0	450

Type of cost: NHS
Description: Short Courses
Item Description: Experience-based co-design course
Justification of Cost: Delivered by The Point of Care Foundation, this 1-day course looks at patient-centred designing and implementation of quality improvements utilising the experiences and views of patients. https://www.pointofcarefoundation.org.uk/event/ebcd-july-2022/

	Year 1	Year 2	Year 3	Total Cost (£)

	0	395	0	395

Type of cost: NHS
Description: Short Courses
Item Description: Facilitator skills training
Justification of Cost: This course will help improve facilitation skills and confidence to facilitate both online and face-to-face events. It will guide how to involve a wide range of voices and ensure everyone feels heard, give ideas for enabling people to work together, get the best out of groups, and explore issues commonly faced by facilitators. This will assist me in the co-creation and focus groups.
https://www.pointofcarefoundation.org.uk/event/one-day-facilitation-course/

	Year 1	Year 2	Year 3	Total Cost (£)
	395	0	0	395

Type of cost: NHS
Description: Short Courses
Item Description: Introduction to Participatory Action Research
Justification of Cost: Delivered by the Social Research Association, this course will teach me how to conduct PAR, the stages that must be followed when designing PAR, and the practicalities.
https://thesra.org.uk/SRA/Shared_Content/Events/Event_display.aspx?EventKey=PAR240822

	Year 1	Year 2	Year 3	Total Cost (£)
	0	220	0	220

Type of cost: NHS
Description: Short Courses
Item Description: The Inclusive Communication Course
Justification of Cost: Delivered by Course Beetle, this course helps people improve their communication with people who have learning disabilities or communication challenges. Each module will identify potential difficulties and consider practical solutions that can be used to support better communication and feelings of confidence. It is run by Speech and language therapists and specialist advisory teachers. This course is intended to help with my co-creation project although it will also benefit my clinical practice. https://coursebeetle.co.uk/courses/icc/

	Year 1	Year 2	Year 3	Total Cost (£)
	0	180	0	180

Type of cost: NHS
Description: Short Courses
Item Description: Conducting a Systematic Review: A Practical Guide
Justification of Cost: Delivered by Cardiff University, this course aims to equip participants with an understanding of the systematic review process and an introduction to the skills necessary to conduct a review. It should build on the skills I learnt by completing a scoping review in my PCAF and will assist me in conducting my own systematic review in Package 1. https://www.cardiff.ac.uk/professional-development/available-training/short-courses/view/conducting-a-systematic-review-a-practical-guide

	Year 1	Year 2	Year 3	Total Cost (£)
	650	0	0	650

Type of cost: NHS
Description: Short Courses
Item Description: Project Management and Research Governance
Justification of Cost: Delivered by the University of Birmingham, this course will explore the ethical dimensions of research, Good Clinical Practice and the essentials for effective project management including pre-project planning, developing realistic time frames and effective time management.
https://www.birmingham.ac.uk/postgraduate/courses/short-courses/mds/project-management-and-researchgovernance

	Year 1	Year 2	Year 3	Total Cost (£)
	0	1130	0	1130

Type of cost: NHS
Description: Short Courses

Item Description: Emerging Clinical Leaders programme
Justification of Cost: Aimed at healthcare professionals moving into more senior roles, this course aims to provide the skills needed to manage a team, build resilience and respond to an ever-changing external environment. https://www.kingsfund.org.uk/leadership-development/courses/emerging-clinical-leaders

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	2634	2634

Type of cost: NHS
Description: Short Courses
Item Description: Educational media and technologies
Justification of Cost: Delivered by Cardiff University, this course aims to provide the skills to design, develop and critically analyse educational media and technologies to enhance the learning-teaching process. https://www.cardiff.ac.uk/study/postgraduate/taught/standalone-modules/medical-education

	Year 1	Year 2	Year 3	Total Cost (£)
	0	1133	0	1133

Type of cost: NHS
Description: Short Courses
Item Description: Assessing learning
Justification of Cost: Also delivered by Cardiff University, this course aims to critically evaluate the purpose, design, application and interpretation of a range of assessment methods available to clinical educators. https://www.cardiff.ac.uk/study/postgraduate/taught/standalone-modules/medical-education

	Year 1	Year 2	Year 3	Total Cost (£)
	0	567	0	567

Type of cost: NHS
Description: Short Courses
Item Description: MRI Safety Officer (MRSO) course
Justification of Cost: Delivered by MRI Safety Matters, this course introduces clinical decision-making and standardised approaches to scanning patients with devices, implants, and foreign bodies. It discusses assessing risks versus benefits for any requested MR study and teaches the steps that can be undertaken to potentially mitigate those risks to a clinically acceptable level. https://www.mrisafetymatters.co.uk/kanalsmrsocourse2024

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	900	900

Type of cost: NHS
Description: Short Courses
Item Description: Introduction to Artificial Intelligence for Radiographers
Justification of Cost: This introductory AI course will form part of the preparatory work for my proposed post-doctoral plan. The course is designed specifically for the radiographic workforce and covers topics like AI's strengths, weaknesses and opportunities, principles of computer science and ethics. https://www.city.ac.uk/prospective-students/courses/professional-development/introduction-to-artificial-intelligence-for-radiographers

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	2360	2360

Type of cost: NHS
Description: Applicant PhD Tuition Fees
Item Description: Tuition fees for the Institute of Cancer Research (ICR)
Justification of Cost: This is the cost of tuition at the ICR

	Year 1	Year 2	Year 3	Total Cost (£)
	4786	4786	4786	14358

	Year 1	Year 2	Year 3	Total Cost (£)
Training And	10777	12171	11846	34794

Development Total				

Total Direct Costs

	Year 1	Year 2	Year 3	Total Cost (£)
Total Direct Costs	109704	171839	97818	379361

Indirect costs

HEI Indirect costs
No costs have been added

Commercial Indirect Costs
No costs have been added

Other Partner Organisation Indirect costs
No costs have been added

	Year 1	Year 2	Year 3	Total Cost (£)
Total Indirect Costs	0	0	0	0

Research Costs Requested From Funder

	Direct costs	Indirect costs/Overheads	Amount requested
Total Higher Education Institution Costs (80%)	0	0	0
Total Higher Education Institution Costs (100%)s	0	0	0
Total Commercial Costs	0	0	0
Total Other Parther Organisation Costs	0	0	0
Total NHS Research Costs	379361	0	379361
Total Research Costs Requested From Funder	379361	0	379361

Summary

	Year 1	Year 2	Year 3	Total Cost (£)
Salaries	**76000**	**76000**	**76000**	**228000**
Travel-subsistence-and-conference-fees	**1000**	**1000**	**1000**	**3000**
Equipment	**990**	**0**	**0**	**990**
Consumables	**389**	**0**	**0**	**389**
Patient-and-Public-Involvement	**19248**	**32876**	**5305**	**57429**
Other-Direct-Costs	**1300**	**49792**	**3667**	**54759**

Training And Development	**10777**	**12171**	**11846**	**34794**
HEI-Indirect-costs	**0**	**0**	**0**	**0**
Commercial-Indirect-Costs	**0**	**0**	**0**	**0**
Other-Partner-Organisation-Indirect-costs	**0**	**0**	**0**	**0**
Total	**109704**	**171839**	**97818**	**379361**

NHS Support Costs

Cost Item:	Year 1	Year 2	Year 3	Total Cost (£)
Informed consent	0	1567	0	1567.0

Justification:
Informed consent : This is the cost of conducting informed consent with radiographer participants before the start of the training evaluation. It covers 20 minutes per participant, with 100 participants in total.

	Year 1	Year 2	Year 3	Total Cost (£)
NHS Support Costs Total	0	1567	0	1567

Have you discussed and agreed these costs with the Lead Network?	Yes

NHS Treatment Costs
No costs have been added

Have you discussed and agreed the NHS costs with the Lead Trust?	N/A

Is the patient/service user care being provided different from the usual standard treatment for the condition?
N/A

Usual Treatment Costs
No costs have been added

Excess Treatment costs/(savings)

	Cost per patient(£)	Year 1	Year 2	Year 3	Total Cost (£)
Total NHS Excess Treatment costs/(savings)	0	0	0	0	0

	Total (£)
NHS Support Costs requested from networks	1567
NHS Excess Treatment costs/savings	0
TOTAL NHS SUPPORT AND TREATMENT COSTS	1567

Total Funding Required (including non-research costs)

	Total (£)
Total Research Costs Requested from Funder	379361
Total NHS (Support and Excess Treatment) Costs	1567
Total Funding Required	380928

10. Uploads

Please upload the following documents, if necessary.

When uploading a document to your application, please ensure that it is in an appropriate file format. It is not possible to upload JPEG images, and with the exception of the SoECAT Form, which should be uploaded as an Excel spreadsheet, all files should be uploaded in Adobe PDF or Word DOC/DOCX format. Please refer to the Guidance Notes for further information on the attachments required.

(References - References.pdf) is included as an appendix within this file.

(Figures / Tables - eLfH example images.pdf) is included as an appendix within this file.

(Research Timetable - Gantt chart.pdf) is included as an appendix within this file.

(SoECAT Template - Funder Export_345198_SoECAT_Educational resources for MRI safety.xlsx) is included as an appendix within this file.

11. Participants and Signatories

Participants
You are required to supply the name and email address (if not already registered on the ARAMIS application system) of each required participant. Everyone included in this section will be acting as a 'participant' in your application and must agree to be named as such.

By confirming participation, participants are acknowledging their involvement and input into this application, and are agreeing to be involved, as indicated, in any resultant fellowship.

Once the 'Administrative Authority or Finance Officer' participant has confirmed their participation, they will be able to access the 'Detailed Budget' section of the application form. You should collaborate with your 'Administrative Authority or Finance Officer participant to ensure that this section is completed satisfactorily.

You must ensure that all participants are happy for your application to be submitted before you submit it.

It is permissible for a single individual to serve as both a signatory and a participant, but they will need to be added separately under each heading.

Named participants must confirm their participation on your application before you will be able to select the 'SUBMIT' option. Please refer to the Submission Process Flow Diagram and the Applicant Guidance Notes for further information.

Primary Doctoral Supervisor

Doctoral Primary Supervisor Participant 1

Title	Professor
Forename(s)	Helen
Surname	McNair
Position	Research Radiographer

Additional Academic Supervisor(s)

Supervisor Participant 1

Title	Dr
Forename(s)	Susanne
Surname	Cruickshank
Position	Strategic Lead for Applied Health Research

Supervisor Participant 2

Title	Dr
Forename(s)	Christina
Surname	Malamateniou
Position	Associate Professor/Reader in Radiography

Practice Supervisor(s)

Practice Support Participant 1

Title	Dr
Forename(s)	Marco
Surname	Borri
Position	Lead Clinical Scientist (Magnetic Resonance)

Practice Support Participant 2

Title	Ms
Forename(s)	Erica
Surname	Scurr
Position	Lead MRI Superintendent Radiographer

Practice Support Participant 3

Title	Dr
Forename(s)	Ian
Surname	Simcock
Position	Clinical-academic radiographer + Senior Research Fellow

Administrative Authority or Finance Officer

Administrative Authority or FO Participant

Title	Ms
Forename(s)	Karry
Surname	Tymieniecka
Position	Associate Finance Director

Signatories
You are required to supply the name and email address (if not already registered on the ARAMIS application system) of each required 'Head of Department / Senior Manager' signatory. Once their contact details have been entered below, these signatories can be invited to log in and confirm their participation in your application.

Once the nominated signatories have confirmed their participation, they can access and complete the required 'Host Organisation Statement of Support' in the 'Training & Development and Research Support' section of the application form.

Your signatories must confirm their participation in your application and complete the section they are responsible for before you can select the 'submit' option.

Signatories must also approve your application after you select the 'submit' option but BEFORE the application submission deadline.

Signatory approval from your 'Head of Department / Senior Manager' signatories will result in the application being fully submitted to the NIHR. All parties (applicant, participants and signatories) will be notified of this via an automated system-generated email.

The NIHR will not accept your application unless fully approved by your signatories before the 1 p.m. deadline on Tuesday 11th June 2024.

Heads of Departments / Senior Managers

Head of Department or Senior Manager 1

Title	Dr
Forename(s)	Susanne
Surname	Cruickshank
Position	Strategic Lead for Applied Health Research

Head of Department or Senior Manager 2

Title	Professor
Forename(s)	Kevin
Surname	Harrington
Position	

12. Acknowledgement, Review and Submit

Conflict checks: Please declare any conflicts or potential conflicts of interest that you may have in undertaking this research, including any relevant, non-personal & commercial interest that could be perceived as a conflict of interest.

By completing this form, the programme reserves the right to share, in confidence, details of your application with other parts of the NIHR and with the Department of Health and Social Care.

Agreement to Terms and Conditions:

I confirm that I have secured all necessary licences and ethical approvals in relation to the research and will abide by the terms of those licences and approvals in carrying out the research.

I confirm that I have read the NIHR Carbon Reduction Guidelines and, where possible, have taken steps to reduce the carbon emissions generated by this research.

I confirm that all material within this application, with the exception of sections that other participants are required to complete, is my own work and I have taken steps to ensure that no parts of the application have been inappropriately plagiarised.

I undertake to conduct the research according to such guidelines for good research practice as the Host Organisation may from time to time lay down according to the NIHR research governance framework.

I undertake to conduct the research according to such guidelines for good research practice as the Host Organisation may from time to time laydown according to the UK Policy Framework for Health and Social Care Research.

In completing this, I declare that, to the best of my knowledge, the information provided in this application, is true, accurate and complete. I have read the Guidance Notes and agree to accept the process by which an application is assessed and agree to abide by the conditions under which an award may be granted.

☒Confirmed

Appendices

1. References.pdf
2. eLfH example images.pdf
3. Gantt chart.pdf
4. Funder Export_345198_SoECAT_Educational resources for MRI safety.xlsx

Appendix #1: References.pdf

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Appendix #2: eLfH example images.pdf

Active implants ▸ Pacemakers and implantable cardioverter-defibrillators

Menu | Previous | 4 of 32 | Next

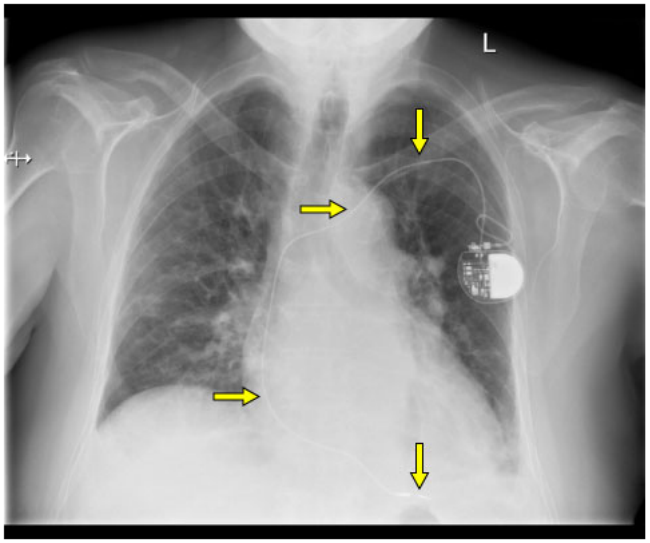
Overview | Important points

Select the hyperlinks below.

A pacemaker is an active implant that sends electrical pulses to the heart to maintain an optimal beating rate. An implantable cardioverter-defibrillator (ICD) is an active implant that monitors heart rate and can perform pacing, cardioversion or defibrillation if a dangerous heart rhythm is noticed [0b]. You can't tell the difference between them on a chest radiograph. However, ICDs are often larger due to the battery size needed to send a defibrillation shock

Traditional style pacemakers and ICDs consist of a generator box commonly located in the axilla, which emits the electrical pulses and pacing leads which connect to the heart (myocardium) and deliver the electrical pulses [1].

Generator boxes and leads will look different depending on make, model and manufacturer. The generator box is most often found in the left axilla, however, it can also be located on the right.



Zoom

*Select the **Important points** tab to continue.*

Passive implants ▸ Patent foramen ovale closure device

Menu | Previous | 22 of 32 | Next

Select the hyperlink below.

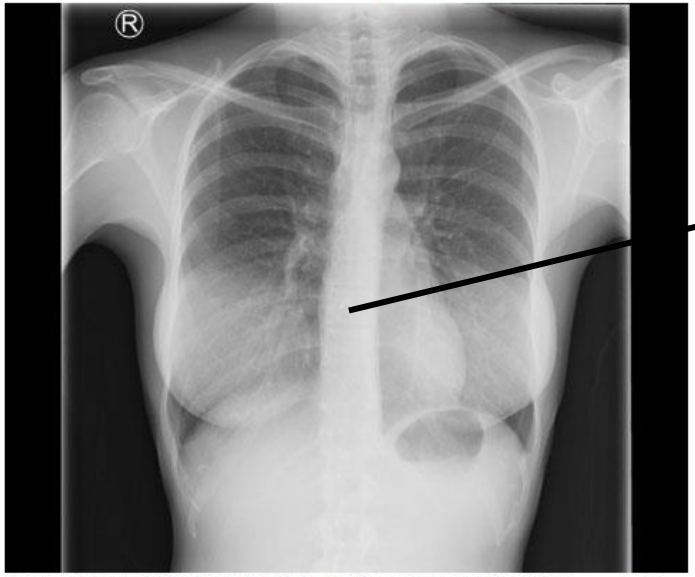
A patent foramen ovale (PFO) closure device is used to treat the incomplete fusion of the patent foramen ovale, between the left and right atrium, which usually occurs postpartum [37].

PFO closure devices commonly appear as two discs joined by a central neck. The size and shape may vary depending on the manufacturer, make and model.

A PFO closure device will be found in the superior portion of the heart, between the left and right atrium.

Important points

- PFO closure devices are placed endovascularly, meaning they cannot be seen externally.



Case courtesy of Dr Bruno Di Muzio, Radiopaedia.org, rID: 74347 Zoom

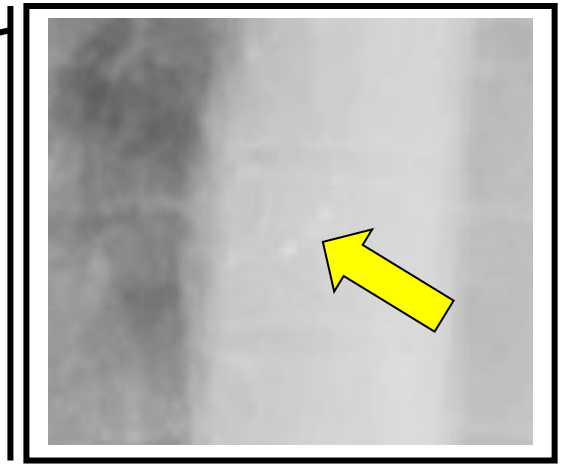
Figure.1

Figures 1 and 2 showcase the type of training material and content that the eLfH eLearning module, which is part of the output of this fellowship, is expected to produce.

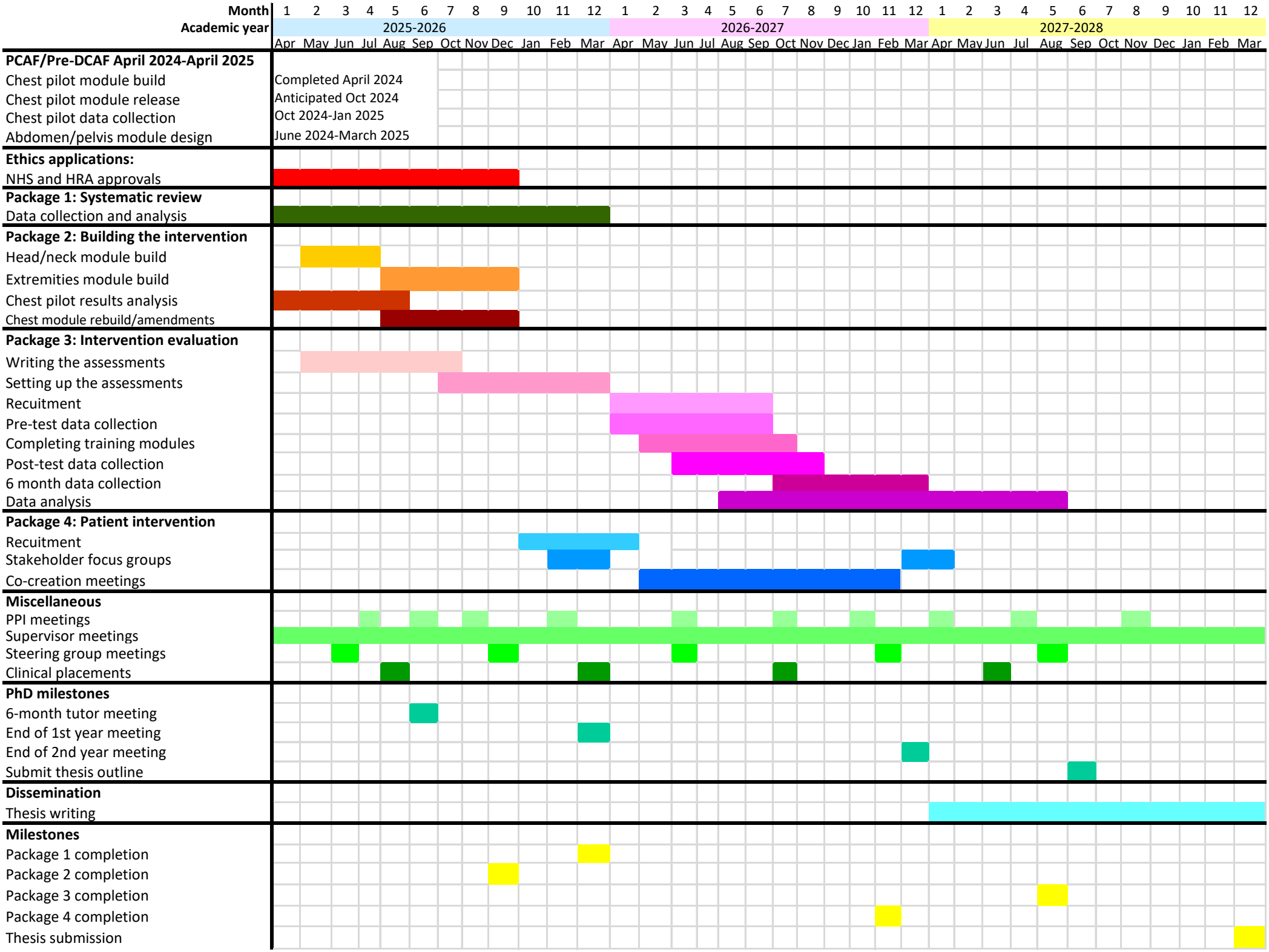
This training module aims to teach radiographers to identify medical implants on X-ray images. Both slides are taken from the current draft of the 'chest implants' module and describe a medically implanted device as seen on X-ray images. The location of the implant (a pacemaker in Figure 1) is identified on the images using graphical aids (e.g., arrows), activated using the hyperlinks in the text.

Where a pacemaker (Figure 1) is quite visible on an X-ray image, Figure 2 demonstrates a Patent Foramen Ovale (PFO) closure device, which is very small and subtle in comparison. Yet, both need to be identified and clinical decisions made before proceeding with the MRI scan. The consequences of not identifying the implants could be a delayed or cancelled scan, or a less desirable modality might be used. Radiographers therefore need to be trained to identify both obvious and more subtle implanted medical devices.

Figure.2



Appendix #3: Gantt chart.pdf


Month
1 2 3 4 5 6 7 8 9 10 11 12 1 2 3 4 5 6 7 8 9 10 11 12 1 2 3 4 5 6 7 8 9 10 11 12
Academic year
2025-2026
2026-2027
2027-2028
Apr May Jun Jul Aug Sep Oct Nov Dec Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec Jan Feb Mar
PCAF/Pre-DCAF April 2024-April 2025
Chest pilot module build
Completed April 2024
Chest pilot module release
Anticipated Oct 2024
Chest pilot data collection
Oct 2024-Jan 2025
Abdomen/pelvis module design
June 2024-March 2025
Ethics applications:
NHS and HRA approvals
Package 1: Systematic review
Data collection and analysis
Package 2: Building the intervention
Head/neck module build
Extremities module build
Chest pilot results analysis
Chest module rebuild/amendments
Package 3: Intervention evaluation
Writing the assessments
Setting up the assessments
Recuitment
Pre-test data collection
Completing training modules
Post-test data collection
6 month data collection
Data analysis
Package 4: Patient intervention
Recuitment
Stakeholder focus groups
Co-creation meetings
Miscellaneous
PPI meetings
Supervisor meetings
Steering group meetings
Clinical placements
PhD milestones
6-month tutor meeting
End of 1st year meeting
End of 2nd year meeting
Submit thesis outline
Dissemination
Thesis writing
Milestones
Package 1 completion
Package 2 completion
Package 3 completion
Package 4 completion
Thesis submission

Appendix #4: Funder Export_345198_SoECAT_Educational resources for MRI safety.xlsx